Labor Market Concerns and Support for Immigration*

Ingar Haaland† Christopher Roth‡

August 14, 2017
This version: August 12, 2018

Abstract

Do labor market concerns drive people's support for immigration? Using a large, representative sample of the US population, we first elicit beliefs about the labor market impact of immigration. To generate exogenous variation in beliefs, we then provide respondents in the treatment group with research evidence showing no adverse labor market impacts of immigration. Treated respondents update their beliefs and become more supportive of immigration, as measured by self-reported policy views and petition signatures. Treatment effects also persist in a follow-up study designed to mitigate experimenter demand. Our results demonstrate that labor market concerns causally affect people's support for immigration. (*JEL* C91, D83, F22, J15)

Keywords: Beliefs about labor markets, support for immigration, political behavior

*We thank Alexander Cappelen, Simon Quinn, and Bertil Tungodden for extremely generous guidance. We also thank seminar audiences in Bergen (University of Bergen, NHH, and ECBE), Berlin, Essex, Lofoten, Munich, New York (NY Fed), Nottingham, Oxford, Richmond (ESA), San Diego, Stanford (SITE), Trondheim, and Warwick for valuable comments. The experiment is registered in the AEA RCT Registry as trial 2247, `https://www.socialscienceregistry.org/trials/2247`. This work was partially supported by the Research Council of Norway through its Centres of Excellence Scheme, FAIR project No 262675. IRB approval was obtained from the University of Oxford.

†NHH Norwegian School of Economics (email: Ingar.Haaland@nhh.no)

‡Institute on Behavior & Inequality (email: Chris.Roth@briq-institute.org)

Electronic copy available at: https://ssrn.com/abstract=3017420

1 Introduction

There is currently a heated debate about immigration in Western countries. Arguments about the adverse labor market effects of immigrants are prominent in this debate, and voters are deeply divided in their beliefs about how immigration affects the labor market outcomes of natives. In the United States, for instance, 57 percent of those in favor of reducing immigration say that immigrants take jobs away from Americans compared to only 23 percent of those not in favor of reducing immigration (Smith et al., 2014).

While this correlation could reflect a causal effect of labor market concerns on attitudes towards immigration, it could also be driven by reverse causality or omitted variable bias. We solve this identification challenge by experimentally manipulating beliefs about the labor market impact of immigration. This allows us to provide the first causal evidence on whether people's beliefs about the labor market impact of immigration are a causal driver of support for immigration.

In a pre-registered experiment with a large and representative sample of the US population, we first elicit beliefs about the labor market impact of the Mariel boatlift, which is known as the "one historical event that has most shaped how economists view immigration" (Clemens, 2017). During the Mariel boatlift, which was an unexpected mass immigration of Cubans to the United States, the low-skilled workforce in Miami increased by 20 percent over the course of a few months. To generate exogenous variation in beliefs, we provide a random subsample of our respondents with information about research showing no adverse labor market impacts of the Mariel boatlift on wages and unemployment in Miami (Card, 1990). We then measure our respondents' support for immigration using both self-reported attitudes on preferred immigration levels as well as behavioral measures: two

Electronic copy available at: https://ssrn.com/abstract=3017420

anonymous real online petitions proposing changes to the annual cap on visas for low-skilled guest workers to the US. Finally, we conduct an obfuscated follow-up study one week later in which we hide the connection between the follow-up and the main study. This allows us to test for robustness of treatment effects and to address concerns about experimenter demand (de Quidt et al., 2018; Zizzo, 2010).

The main finding of the paper is that beliefs about the labor market impact of immigration are an important causal driver of people's support for immigration. Treated respondents update their beliefs when provided with research evidence showing no adverse labor market impacts: the treatment changes people's beliefs about the labor market impact of low-skilled immigration by 0.16 of a standard deviation in the direction of more positive impacts ($p<0.01$). This exogenous shift in beliefs leads to increased support for immigration as measured by both self-reports and real online petitions. Treated respondents increase their support for admitting more low-skilled immigrants by 0.14 of a standard deviation ($p<0.01$). This effect size corresponds to about one quarter of the Democrat–Republican difference in policy views. Treated respondents are also 2.7 percentage points more likely to sign a real online petition in favor of increasing the annual cap on low-skilled guest workers to the US, which corresponds to a 69.2 percent increase from the control group mean of 3.9 percentage points ($p<0.01$). Moreover, we show robustness of treatment effects over time. In the one-week follow-up, we find statistically significant treatment effects that are not statistically significantly different from the ones observed in the main study.

Exploring heterogeneity in treatment responses, we find that respondents with pessimistic pre-treatment beliefs about the labor market impact of immigration become more optimistic and vice versa. We confirm these patterns using techniques from machine learning (Athey and Imbens, 2016), which show that people's pre-

Electronic copy available at: https://ssrn.com/abstract=3017420

treatment beliefs about the wage impact of the Mariel boatlift are the best predictor of heterogeneous responses to the treatment. Further exploring mechanisms, we find that the treatment does not significantly affect beliefs unrelated to the labor market, such as beliefs about the cultural impact of immigration. These results corroborate our interpretation that the treatment operates through changes in beliefs about the labor market impact of immigration.

Our work contributes to several strands of the literature. First, we contribute to the literature on labor market concerns and attitudes towards immigration (Card et al., 2012; Dustmann and Preston, 2001, 2006; Facchini et al., 2009; Hainmueller and Hopkins, 2014a; Hainmueller and Hiscox, 2010; Hainmueller et al., 2015; Iyengar et al., 2013; Mayda, 2006; Scheve and Slaughter, 2001). A seminal study by Citrin et al. (1997) used data from the American National Election Studies to explore the determinants of people's attitudes towards immigration. The study found that while personal economic circumstances do not predict attitudes towards immigration, beliefs about whether immigrants take jobs away from natives and cause higher taxes were robustly predicting support for immigration.[1] However, as stressed by the authors, the correlations between beliefs and policy views are vulnerable to omitted variable bias and reverse causality.

To overcome the challenges with observational data, researchers have used experiments which measure support for hypothetical immigrants with randomly assigned characteristics, such as their education levels and whether they plan to find work. Such experiments can causally identify which characteristics of immigrants people value, but do not allow us to identify the underlying motivations for why certain characteristics are valued. For instance, an experiment by Hainmueller and Hopkins (2014a) finds that Americans are concerned about low-skilled immigrants

[1] Subsequent studies have debated whether personal economic circumstances predict support for immigration (Hainmueller et al., 2015; Mayda, 2006; Scheve and Slaughter, 2001).

Electronic copy available at: https://ssrn.com/abstract=3017420

who do not plan to find work. This finding could reflect economic considerations about the fiscal burden posed by these immigrants, but it could also reflect concerns about how low-skilled immigrants who do not plan to find work fit in culturally in the US. Similarly, an experiment by Sniderman et al. (2004) finds that cultural characteristics evoke more opposition to immigration than economic characteristics. This has been taken as evidence that opposition to immigration is primarily rooted in cultural concerns. However, a challenge to this interpretation is that cultural characteristics, such as not speaking the local language, may be valued because of their economic implications.

We employ a different approach by experimentally manipulating our respondents' beliefs about the economic impact of immigration. Since we find that our experimental manipulation does not significantly affect our respondents' beliefs about the cultural impact of immigration, we are able to cleanly identify the role of economic concerns in driving support for immigration. In contrast to previous experimental work which finds a seemingly limited role of economic concerns in driving support for immigration, we find that economic concerns are a quantitatively important causal driver of attitudes towards immigration.

We also contribute to the literature that tries to understand the determinants of people's policy preferences by experimentally manipulating beliefs (Alan and Ertac, 2017; Alesina et al., 2018b; Cruces et al., 2013; Gilens, 2001; Karadja et al., 2017; Kuklinski et al., 2000; Kuziemko et al., 2015). This literature has mostly focused on correcting biases in people's beliefs about verifiable facts, for instance related to income inequality. We differ from these studies by shifting people's beliefs about the causal relationship between two economic variables which inform the functioning of markets. Our approach of eliciting people's beliefs about the outcomes of natural experiments allows us to create exogenous variation in beliefs by assigning some

Electronic copy available at: https://ssrn.com/abstract=3017420

respondents to research evidence on the same natural experiments.

In the context of immigration policy, our results differ from previous experiments showing muted responses of policy preferences to factual information about the fraction of immigrants (Hopkins et al., 2018; Sides and Citrin, 2007), or their characteristics (Alesina et al., 2018a; Grigorieff et al., 2018; Lergetporer et al., 2017). It has been suggested that most policy preferences are hard to move because people are unable to "connect their concerns on a given issue with the public policies meant to address it" (Kuziemko et al., 2015). A potential explanation for why we find stronger responses to new information than most previous studies could be that we give information that is particularly easy to connect with public policy. Consistent with this explanation, online experiments in Japan show that information about how immigration helps solve social and economic problems is effective in changing attitudes towards immigrants (Facchini et al., 2016).

More broadly, our findings contribute to a long-standing debate in the social sciences which discusses the relative importance of consequential, ideological, and social motives in driving people's political behavior (Bursztyn et al., 2016; DellaVigna et al., 2016; Downs, 1957; Gerber et al., 2009). Our results support a consequential view on political behavior by highlighting that changes in beliefs about the economic consequences of a policy can significantly affect political behavior.

2 Experimental design and sample

Our experiment has two parts: A main experiment and an obfuscated follow-up study performed seven days after the main experiment. In the following, we describe the structure of the main experiment and the obfuscated follow-up study. Figure A.1

Electronic copy available at: https://ssrn.com/abstract=3017420

provides a summary of the structure.[2]

2.1 Main experiment

In the main experiment we first ask questions about demographics, political affiliation and self-perceived skill levels, as well as eliciting our respondents' pre-treatment beliefs about the labor market impact of immigration. We then expose half of our respondents to the information treatment. Subsequently, we measure our respondents' support for immigration using self-reported policy views and signatures on real online petitions. Finally, we elicit post-treatment beliefs about the labor market impact of immigration.

2.1.1 Pre-treatment beliefs about the Mariel boatlift

We first elicit our respondents' beliefs about the labor market impact of the Mariel boatlift, which has strongly shaped how economists view immigration (Clemens, 2017). To familiarize our respondents with the context, we present them with the following text:

> In 1980, Cuba's then President, Fidel Castro, suddenly announced that Cubans wishing to emigrate to the United States were free to do so. This led to an unexpected mass immigration to Miami, Florida, where most of the Cuban immigrants arrived by boat.
>
> With the arrival of the new Cuban immigrants, Miami's workforce grew by 55,000, or 8 percent, almost at once. The new immigrants were mostly

[2]Full instructions for the main experiment and the obfuscated follow-up are provided in Section E and F of the Online Appendix, respectively. The Qualtrics survey for the main experiment is available on the following link: `https://nhh.eu.qualtrics.com/jfe/form/SV_8Am0WWUZiq4u2ax`. The Qualtrics survey for the obfuscated follow-up is available on the following link: `https://cessoxford.eu.qualtrics.com/jfe/form/SV_d71YFolo6Dw9Ump`.

Electronic copy available at: https://ssrn.com/abstract=3017420

> low-skilled, which meant that the low-skilled workforce increased by 20 percent.
>
> The large, unexpected addition of 55,000 new immigrants to the Miami workforce has allowed researchers to study the impact of immigration on the labor market. To do so, the researchers studied wage and unemployment changes in Miami after the mass immigration relative to other US cities that, because of geographic distance, were not affected by the mass immigration of Cubans.

Thereafter, we ask our respondents how they think "the mass immigration of Cubans" affected wages and unemployment in Miami for both low- and high-skilled workers. To minimize the risk that respondents would try to google the answer, we choose not to incentivize these beliefs and avoided to explicitly mention the term "Mariel boatlift" in the instructions.

2.1.2 Research evidence on the Mariel boatlift

Following the belief elicitation, we inform respondents in the treatment group about the results from a seminal study about the labor market consequences of the Mariel boatlift (Card, 1990).[3] Specifically, we present the following text to respondents in the treatment group (Figure A.14 provides a screenshot):

> The researchers who analyzed the short- and long-term effects of the mass immigration of Cubans to Miami concluded that, for both high-skilled

[3] Card (1990) provides compelling evidence on the labor market consequences of the Mariel boatlift by conducting a difference-in-differences analysis. The findings by Card (1990) that the Mariel boatlift did not have negative labor market impacts are consistent with a large body of economic research on the labor market consequences of immigration (Card, 2005, 2012; Dustmann et al., 2005). However, the labor market impact of immigration is still being debated (Borjas, 2014; Card and Peri, 2016).

Electronic copy available at: https://ssrn.com/abstract=3017420

> and low-skilled workers, the mass immigration had **virtually no effect on wages** and **virtually no effect on unemployment**.
>
> According to the researchers, the mass immigration had virtually no effect on wages and unemployment because the new Cuban immigrants increased the overall demand for goods and services, which created more jobs.

Respondents in the control group do not receive any information and proceed directly from the belief elicitation questions to the outcome questions.

2.1.3 Measuring support for immigration: self-reported policy views

To measure how the treatment affects support for immigration, we first investigate self-reported attitudes. Although we give people information about the labor market impact of low-skilled immigration, respondents could also use this information to update their beliefs about the labor market impact of high-skilled immigration. We therefore ask questions about both *low-skilled* and *high-skilled* immigrants. Furthermore, it is possible that the causal effect of beliefs on attitudes depends on the immigrants' cultural characteristics. To fix beliefs about the immigrants' cultural characteristics, we also differentiate between immigrants who are *highly familiar* and *not familiar* with American values and traditions.[4] All respondents are asked about all four types of immigrants, but we randomize the order of the questions between respondents. For example, to measure our respondents' views on low-skilled immigrants not familiar with American values, we administer the following question:

> Immigrants to the US differ in terms of their professional skill levels as well as their familiarity with American values and traditions.

[4]These questions were also used to assess attitudes towards immigration in Hainmueller et al. (2015).

Electronic copy available at: https://ssrn.com/abstract=3017420

> Do you think the US should allow more or less **low-skilled** immigrants that are **not familiar** with American values and traditions to come and live here?

For each immigrant type, respondents report their answer on a 5-point Likert scale from (1) "Allow a lot less of these immigrants" to (5) "Allow a lot more of these immigrants." Section A in the Online Appendix provides a conceptual framework on how people report their policy views on immigration and how the treatment affects self-reported policy views.

2.1.4 Measuring support for immigration: political behavior

After measuring self-reported attitudes, we give our respondents the opportunity to engage in political behavior using real online petitions. We inform our respondents that Congress is debating whether to change the annual cap on non-agricultural guest workers to the US, the H-2B visa program. We chose to focus on the H-2B visa program because it was heavily debated in Congress at the time of the experiment and because of the close connection between our informational treatment, which highlighted the labor market impact of low-skilled immigration, and the H-2B visa program, which is a program to bring low-skilled foreign nationals to the US.

To make sure that the debate surrounding the H-2B visa program is meaningful to the respondents, we suggest some arguments in favor of both increasing and decreasing the annual cap. Respondents are then told that they will be given the opportunity to sign one of two petitions related to this debate. The first petition suggests to increase the annual cap from 66,000 to 99,000, whereas the second petition suggests to decrease the annual cap from 66,000 to 33,000. We randomized the order of the petitions between participants. We chose to create petitions with concrete policy proposals to maximize external validity.

Electronic copy available at: https://ssrn.com/abstract=3017420

We ask our respondents whether they want to sign one of the two petitions. Respondents who say that they want to sign one of the petitions are provided with a link to a real petition that we created on the White House web page, petitions.whitehouse.gov (Figure A.10 provides a screenshot). To identify treatment differences in actual signatures, we provide respondents in the treatment and control group with different links to identical petitions.

Petitions on the White House web page have some noteworthy features. First, our petitions never became public and could only be reached through the links provided in our experiment. This was important to avoid contamination by people from outside the experiment. Second, the White House requires an email confirmation for petitions to count. This makes signings more costly than on petition pages that do not require any form of verification since respondents have to log on to their email account to sign the petition. Third, the petition signatures were anonymous meaning that only the White House could observe the names and emails of the signatories. This mitigates concerns about experimenter demand because respondents cannot use the petitions to signal that they conform to the experimenter's wishes. Fourth, it takes several hours for the petition pages to update the number of signatures. Eventual differences in the number of signatures between the treatment and control petition page could be a confound as people may be more likely to sign a petition which already has more signatures. We measured the number of signatures over time and do not find that the treatment effects get stronger after the number of signatures on the petition pages gets updated (results available upon request).

2.1.5 Post-treatment beliefs about the impact of immigration

To explore mechanisms and to confirm that we successfully managed to induce exogenous variation in beliefs, we examine people's perceptions about how increasing

Electronic copy available at: https://ssrn.com/abstract=3017420

the number of low-skilled or high-skilled immigrants to the United States would affect labor market outcomes and other theoretically relevant dimensions over the next five years. For this block of questions, we randomize whether respondents answer the questions about low-skilled or high-skilled immigrants in order to reduce the risk of survey fatigue.

Since people's views on immigration could be formed by self-interested concerns about how immigration affects their own household as well as broader concerns about how it affects the nation as a whole, we elicit beliefs about both the impact of immigration on both the respondents' *own household* and on *most Americans*. To assess whether the treatment shifted beliefs about the labor market impact of immigration, we ask respondents how they think increased immigration affects wages and job opportunities and job security. To assess whether the treatment changed beliefs not related to the labor market, we also ask how they think increased immigration affects taxes and how it affects American culture and society as a whole. Finally, to examine whether belief updating about the economic impact of immigration depends on the immigrants' cultural characteristics, we ask separate questions about the the overall economic effect of increased immigration of immigrants *not familiar* and *highly familiar* with American values and traditions.

2.2 Obfuscated follow-up study

About one week after the main study, we perform a follow-up survey with the same participants to test for the robustness of treatment effects.[5] We chose to have about

[5]The actual number of days between the main study and the follow-up study varied between one and fourteen days for all subjects. The average difference was seven days. The main reason days vary between respondents it that some respondents choose to take the survey several days after receiving the initial invitation. Furthermore, the market research company sent out invitations to the follow-up study one day after all respondents had completed the main study.

Electronic copy available at: https://ssrn.com/abstract=3017420

one week between the follow-up and the main study to strike a balance between testing for persistence and minimizing attrition.

A potential concern with evidence from our main experiment is that treatment effects could be biased due to experimenter demand effects (Zizzo, 2010). While recent evidence suggests that the bias is not quantitatively important (de Quidt et al., 2018; Mummolo and Peterson, 2018), we take additional steps to address concerns about experimenter demand. Specifically, we obfuscate the purpose of our follow-up study and present it as an independent study to participants. Since no treatment is administered in the follow-up study, differential experimenter demand between the treatment and control group is unlikely to be a concern unless respondents nonetheless realize that the follow-up is connected to the main study.

We take several steps to hide the connection between the main study and the follow-up study from the respondents. First, we choose to collaborate with a market research company where respondents regularly receive invitations to participate in surveys. When sending out these invitations, the market research company uses generic invitation forms that only contain information about pay and expected completion time (Figure A.11 provides a screenshot). Second, we use different consent forms for the two studies: In the first study, respondents are forwarded to a survey with a consent form from the Norwegian School of Economics, while the second presents a consent form from the University of Oxford (Figure A.12 and Figure A.13 provide screenshots). We also use different layouts for the two surveys. Third, to make the follow-up seem like an independent study, we first ask respondents a series of questions about their demographics. Fourth, to further obfuscate the purpose of the follow-up study, we ask several questions about government spending, taxation, and redistribution before we ask any questions about immigration.

At the end of the follow-up study we ask three questions about immigration

Electronic copy available at: https://ssrn.com/abstract=3017420

regarding support for low-skilled immigration, support for high-skilled immigration, and beliefs about the labor market impact of immigration. Since three questions about immigration may send a signal that we are interested in immigration, thus increasing the chance that respondents realize that the two studies are connected, we ask each question on a separate page with the most important outcome question (preference for low-skilled immigration) on the first of these three pages. To further minimize the chance that respondents realize the relationship between the two studies, we use different wordings for the questions on immigration in the follow-up compared to the main study.[6] This also mitigates the concern that treatment effects persist because of people's taste for consistency in survey responses (Falk and Zimmermann, 2012).

While follow-up studies are a common tool in information experiments to test whether treatment effects persist (Alesina et al., 2018b; Facchini et al., 2016; Grigorieff et al., 2018; Kuziemko et al., 2015), our approach of hiding the connection to the main experiment extends the use of follow-up studies to more directly address concerns about experimenter demand. We believe that obfuscated follow-up surveys will be a useful tool for survey experimenters who aim to minimize concerns about experimenter demand effects.

2.3 Sample

We recruited respondents using Research Now, an online market research company in the US that is regularly used by researchers to conduct academic studies (e.g.,

[6]For instance, in the main study, we ask respondents the following question: "Do you think the US should allow more or less low-skilled immigrants that are not familiar with American values and traditions to come and live here?" In the follow-up study, we drop the distinction about familiarity with American values and traditions and simply ask: "In your view, should immigration of workers with little to no education be kept at its present level, increased, or decreased?"

Electronic copy available at: https://ssrn.com/abstract=3017420

de Quidt et al., 2018). We recruited 3130 respondents who are broadly representative of the adult US population on some important observable characteristics. All respondents who had finished the main study were invited to participate in the obfuscated follow-up study, for which we received 2075 respondents. The experiment was run in late May and early June 2017. We submitted a pre-analysis plan to the AEA RCT Registry before we collected any data for the experiment. We pre-specified the sample size, empirical specifications, and our hypotheses in a pre-analysis plan available at: `https://www.socialscienceregistry.org/trials/2247`.

2.4 Summary statistics and balance

In the main experiment the median age is 47; the median income is $72,871; 78 percent of respondents have at least a two-year college degree; 48 percent are male; and 45 percent are in full-time work (as illustrated in Table A.1). Both our main sample and the follow-up sample are representative of the adult US population in terms of some important observables (age, region, gender, and household income; see Table A.2). Furthermore, treatment and control group are balanced in terms of observables both in the main study and the follow-up (Table A.3 and Table A.4) and there is no differential attrition in the response rates to the follow-up (Table A.5). Since there is some selective attrition in the follow-up unrelated to the treatment status, but related to some demographics (for details, see Table A.5), we include controls in all specifications analyzing the follow-up results.[7]

[7] Even though we include controls in all specifications for the analysis of the follow-up results, results without controls are almost identical and, if anything, slightly stronger.

Electronic copy available at: https://ssrn.com/abstract=3017420

3 Results

This section presents our main results. Since our treatment was tailored to shift beliefs about the labor market impact of low-skilled workers, we choose to only collect behavioral measures of support for low-skilled immigrant workers. Throughout the section, our main focus is therefore on support for low-skilled immigration. For the self-reported outcomes, we also report and discuss results for high-skilled immigrants. While our results on self-reported policy views and petition signatures were pre-specified, we also report some results that were not pre-specified, but which naturally follow from the pre-analysis plan. This includes the results presented in Section 3.2 and Section 3.7.1 on belief updating, the IV results in Section 3.5, and the machine-learning approaches to heterogeneous treatment effects in Section 3.7.4. In Section B of the Online Appendix, we discuss a few minor deviations from the pre-analysis plan.

3.1 Pre-treatment beliefs about the Mariel boatlift

To aid interpretation of our main results, we first investigate which pre-treatment beliefs our respondents hold about the labor market impact of the Mariel boatlift. Figure 1 shows the distribution of these beliefs. The large majority of our respondents thinks that the boatlift negatively affected wages and unemployment for low-skilled workers in Miami. By contrast, the large majority of our respondents thinks the boatlift had no effect on wages and unemployment for high-skilled workers in Miami. Since most of the Cuban immigrants were low-skilled, these results suggest that our respondents believe that immigration mainly affects labor market outcomes for native workers with similar skill levels as the immigrants.

Electronic copy available at: https://ssrn.com/abstract=3017420

[Insert Figure 1 here]

We find that our respondents' pre-treatment beliefs about the Mariel boatlift vary systematically by their background characteristics. Differences in beliefs between self-identified Republicans and Democrats are especially pronounced. For beliefs about the impact on low-skilled wages in Miami, the gap between Republicans and Democrats corresponds to approximately one quarter of a standard deviation. We also observe significant correlations between pre-treatment beliefs and college education, race, work status, age, and income (Figure A.2).

3.2 Do beliefs respond to the treatment?

To investigate whether the treatment successfully affects our respondents' beliefs about the labor market impact of immigration, we estimate the following equation using OLS:[8]

$$y_i = \alpha_0 + \alpha_1 \mathrm{T}_i + \boldsymbol{\alpha}_2 \mathbf{x}_i + \varepsilon_i$$

where y_i is the outcome of interest; T_i is an indicator for whether subject i received the research evidence; $\mathbf{x}_i$ is a vector of pre-specified controls[9]; and ε_i is an individual-specific error term. We use robust standard errors for all specifications.

[Insert Table 1 here]

Table 1 presents regression results for post-treatment beliefs about the labor market

[8] All of the results in this paper are robust to employing ordered response models instead of OLS (results available upon requests).

[9] The controls, which were pre-specified, include gender, age, ethnicity, region, household size, household income, education, employment status, party affiliation, whether the respondent was born in the US, whether the subject's parents were born in the US, self-perceived skill-level, and pre-treatment beliefs about the labor market impact of low-skilled (or high-skilled) immigration. The controls are coded as described in the pre-analysis plan. As pre-specified, we also report results without controls in the tables.

Electronic copy available at: https://ssrn.com/abstract=3017420

impact of increased immigration on most Americans. In the regressions, we z-score all outcomes using the mean and standard deviation of the control group. Columns 1 to 3 show our first main result:

Result 1. *The treatment increases people's optimism about the labor market impact of low-skilled immigration by 0.16 of a standard deviation (p<0.01). This effect size corresponds to approximately one-half of the Democrat–Republican difference in beliefs.*

The treatment also increases people's optimism about the labor market of immigration of high-skilled immigration by 0.23 of a standard deviation (p<0.01), and this effect is not significantly different from the treatment effect on beliefs about low-skilled immigrants (p=0.32). The similar size of treatment effects could reflect that we gave treated respondents a reason for why the Mariel boatlift did not adversely affect labor market outcomes, namely that "the immigrants increased the overall demand for goods and services, which created more jobs," which is not specific to low-skilled immigrants.

Did the treatment affect beliefs about the labor market impact on the respondents' own household? We find that the treatment mostly did not affect our respondents' beliefs about the labor market impact of immigration on their own household (columns 1 and 3 of Table A.6). The only exception is that the treatment increases optimism about the impact of high-skilled immigrants on job security for the respondents' own household by 0.16 of a standard deviation (p<0.01). That we observe larger treatment effects for beliefs about most Americans could reflect that most control group respondents are more optimistic about the labor market impact of immigration on their own household than on most Americans (Figure A.3). This finding is consistent with a large literature in psychology documenting that people form too optimistic beliefs about their own situation relative to others (Kruger and Dunning,

Electronic copy available at: https://ssrn.com/abstract=3017420

1999). Previous literature has suggested that beliefs about the labor market impact of immigration may affect beliefs about (i) the fiscal burden of immigration and (ii) the cultural impact of immigration (e.g., because immigrants who find work are complying with American work-related norms). We find that the treatment did not significantly affect beliefs about the fiscal burden of immigration or beliefs about the cultural impact of immigration (columns 5–7 of Table A.6).

Overall, we find a significant effect of our informational treatment on beliefs about the labor market impact of immigration. Furthermore, we do not find evidence that the treatment significantly affected beliefs not related to the labor market. This allows us to investigate whether changes in beliefs about the labor market impact of immigration have a causal impact on support for immigration.

3.3 Do policy preferences respond to changes in beliefs?

Table 2 presents regression results for our main outcome measures on self-reported policy views. We use a pre-specified index to assess self-reported policy views on low-skilled immigration. The index is defined as the average of the standardized responses to the following two questions: support for increasing the number of low-skilled immigrants that are (i) *highly familiar* with American values and traditions and *not familiar* with American values and traditions. Column 3 shows our second main result:

Result 2. *The treatment increases support for low-skilled immigration by 0.14 of a standard deviation (p<0.01). This effect size corresponds to one fifth of the Democrat–Republican gap in policy preferences.*

[Insert Table 2 here]

Electronic copy available at: https://ssrn.com/abstract=3017420

Column 5 shows that the treatment also increases support for high-skilled immigration by 0.07 of a standard deviation ($p<0.05$). The treatment effect on support for low-skilled immigrants is significantly larger than the effect on high-skilled immigrants ($p<0.01$). Given that we observe a similar effect on beliefs about the labor market impact of high-skilled and low-skilled immigrants, one interpretation of this result could be that labor market concerns are less important for people's attitudes towards high-skilled immigrants; perhaps people think high-skilled workers are more able to cope with reduced wages or less safe job prospects. Alternatively, it could be that the treatment simply had less scope to change attitudes towards high-skilled immigrants as control group respondents are about one quarter of a standard deviation more supportive of high-skilled immigration than low-skilled immigration.

Do treatment effects depend on the familiarity of immigrants with American values and traditions? For both low-skilled and high-skilled immigrants, we observe significantly larger treatment effects for immigrants that are not familiar with American values and traditions (both $p<0.05$; see columns 1–2 and 4–5). These results are consistent with stronger changes in beliefs about the overall economic impact of immigrants *not familiar* with American values compared to immigrants who are *highly familiar* with these (Table A.7).

Does the treatment mainly increase support for immigration or mainly decrease opposition to immigration? Figure 2, which shows the distribution of responses for both control and treatment group respondents, highlights that the treatment makes people both less likely to support decreased immigration and more likely to support increased immigration. For instance, the treatment increases the share who say that they want to "allow more" or "allow a lot more" of low-skilled immigrants that are not familiar with American values and traditions by 29 percent and decreases the

Electronic copy available at: https://ssrn.com/abstract=3017420

share saying that they want to "allow less" or "allow a lot less" by 15.7 percent (see also Figure A.4).

3.4 Behavioral measures

To examine whether changes in self-reported attitudes also translate into changes in political behavior, we analyze whether the treatment affects people's willingness to sign real online petitions regarding the cap on non-agricultural guest workers to the US, the H-2B visa program.

[Insert Table 3 here]

First, we analyze how the treatment affects respondents' stated willingness to sign one of the petitions. Table 3 presents the results.[10] Column 1 shows that the treatment increases the share of respondents who say they would sign the petition to increase the annual cap on H-2B visas by 4.8 percentage points ($p<0.01$), which corresponds to a 16.7 percent increase from the control group mean of 28.6 percentage points. Similarly, column 2 shows that the treatment decreases the share of respondents who say they want to sign the petition to decrease the annual gap by 6.1 percentage points ($p<0.01$), corresponding to a 18.9 percent decrease from the control group mean of 32.2 percentage points.

Next, we investigate whether the observed changes in intentions to sign the petitions are reflected in actual petition signatures. Column 4 demonstrates that this is the case for the petition suggesting an increase in the annual cap:

Result 3. *The treatment increases the share actually signing the petition in favor of*

[10] In this section, we focus on the results without controls since controls are not available for actual signings of the petitions. However, including controls for the intentions to sign the petitions yields identical results.

Electronic copy available at: https://ssrn.com/abstract=3017420

increasing the annual cap by 2.7 percentage points (p<0.01). This corresponds to a 69.2 percent increase from the control group mean of 3.9 percentage points.

Column 5 shows that the treatment decreases the share actually signing the petition in favor of reducing the annual cap by 0.4 percentage points, which corresponds to a 8.8 percent decrease in signatures from the control group mean of 4.5 percentage points. This difference, however, is not statistically significant from zero (p=0.57). One possibility as to why intentions and actual behavior differ for this petition could be that the treatment was less effective in changing beliefs for respondents at the margin of actually signing the petition than for respondents at the margin of intending to sign the petition.[11]

Our results illustrate the importance of collecting data on real behavior in addition to intentions. We find large and significant treatment effects on intentions to sign both petitions, but comparing actual signatures we only find large and significant treatment effects on one of the petitions. Overall, it is encouraging that the treatment effect carried over from intentions to actual behavior for the petition suggesting to increase the annual cap on H-2B visas. This result demonstrates a causal role of labor market concerns on people's political behavior.

3.5 Interpreting effect sizes: IV

What do our results imply about the quantitative importance of labor market concerns in driving support for low-skilled immigration? To answer this question, we examine how beliefs about the labor market impact of immigration affects people's self-reported support for low-skilled immigration by using the treatment assignment as

[11]For a discussion on how treatments effects in information experiments depend on the density of respondents at the margin of taking an action and how much the beliefs of those at the margin are moved, see Coffman et al. (2015).

Electronic copy available at: https://ssrn.com/abstract=3017420

an instrument for beliefs. Specifically, we estimate the following IV regression:

$$y_i = \beta_0 + \beta_1 \widehat{\text{labor_concerns}}_i + \boldsymbol{\beta}_2 \mathbf{x}_i + \varepsilon_i$$

where the first-stage equation is

$$\text{labor_concerns}_i = \gamma_0 + \gamma_1 \text{T}_i + \boldsymbol{\gamma}_2 \mathbf{x}_i + \epsilon_i$$

and where y_i is support for low-skilled immigration; T_i is the treatment indicator; labor_concerns_i is an unweighted index of people's beliefs about the effect of low-skilled immigration on (i) wages for most Americans and (ii) job opportunities or job security for most Americans; and $\mathbf{x}_i$ is a vector of controls.

To satisfy monotonicity of the instrument, we exclude respondents who pre-treatment thought that the Mariel boatlift had either positive or no effects on wages and unemployment in Miami.[12] The first-stage F statistic with the restricted sample is above 20, which confirms instrument relevance. The exclusion restriction—that the treatment only affects support for low-skilled immigration through changes in beliefs about its labor market impact—cannot be directly tested. However, as discussed in Section 3.2, we do not find evidence that the treatment significantly affected beliefs not related to the labor market.[13]

[Insert Table 4 here]

Table 4 reports the IV results for the subsample of respondents who had pes-

[12]Respondents who thought that the Mariel boatlift had a positive labor market impact should become less in favor of immigration after receiving the research evidence (Section A).

[13]The exclusion restriction also rules out that people develop more positive feelings towards immigrants as a result of the treatment. While we cannot test this assumption with our data, we argue that concerns about temporary changes in, e.g., empathy towards immigrants are mitigated in the follow-up study.

Electronic copy available at: https://ssrn.com/abstract=3017420

simistic pre-treatment beliefs about the labor market impact of the Marial boatlift. Columns 1 and 2 show the reduced form effects of the treatment on beliefs and support for low-skilled immigration using the restricted sample. These results from the restricted sample are quantitatively similar to the results from the whole sample. Column 3 of Panel A illustrates our fourth main result:

Result 4. *A one standard deviation change in beliefs about the labor market impact of low-skilled immigration translates into a 0.62 standard deviation change in support for low-skilled immigration (p<0.01).*

This result demonstrates a quantitatively important role for labor market concerns in driving support for low-skilled immigration.[14] As a robustness check, Panel B of Table 4 shows that the IV results are barely affected when we control for post-treatment beliefs about (i) the impact of immigration on wages and employment opportunities of the respondents' own household, (ii) the tax burden created by immigrants for most Americans and the respondents' own household, and (iii) the cultural impact of immigration.

How externally valid are our IV estimates? To shed light on this question, we compare our IV estimates to the corresponding OLS estimates in which we regress support for low-skilled immigrants on our measure of labor market concerns, labor_concerns_i (Column 4 of Table Table 4). The OLS estimates are statistically significant in the same direction as the IV estimates, but are only about half as large in magnitude. One reason for this could be that possible confounding factors in the OLS regressions, such as measurement error in beliefs, are downward biasing the estimated relationship between labor market concerns and support for immigration. Alternatively, it could be that the IV estimates are higher because they capture the local average treatment effect for the subsample of respondents who change their

[14] IV results for high-skilled immigration are similar (as shown in Table A.8).

Electronic copy available at: https://ssrn.com/abstract=3017420

beliefs about the labor market impact of immigration after receiving the research information. It is possible that these respondents have a weaker ideological stance on immigration and are therefore more willing than others to change their policy views.

3.6 Do changes in attitudes persist in the obfuscated follow-up?

Did our information treatment genuinely shift beliefs and policy preferences? Column 7 of Table 2 shows our fifth main result:

Result 5. *The treatment effect on support for low-skilled immigration persists in the obfuscated follow-up. The treatment effect corresponds to 0.10 of a standard deviation (p<0.01)*

We also see some persistence of treatment effects for high-skilled immigrants (column 8 of Table 2), but the point estimate of 0.06 of a standard deviation is not statistically significant (p=0.13). It is also not statistically different from the effect on support for low-skilled immigration (p=0.43). Column 5 of Table 1 shows persistent treatment effects on beliefs about the overall labor market impact of immigration (p<0.01).[15]

Since we use different questions in the follow-up and the main study, the treatment effects observed in the follow-up are not directly comparable to the treatment effect from the main study. Nevertheless, comparing magnitudes, the effect sizes we observe on support for low-skilled and high-skilled immigrants in the follow-up are not significantly different from the effect sizes observed in the main study (p=0.44

[15] We explored whether there was any heterogeneous treatment responses by number of days between the main study and the follow-up, but did not find any systematic differences based on this measure.

Electronic copy available at: https://ssrn.com/abstract=3017420

and p=0.94, respectively).[16]

Overall, we believe the results from the obfuscated follow-up demonstrate that our respondents genuinely changed their attitudes and beliefs about immigration and that it is unlikely that demand effects or emotional responses to the treatment were severely biasing the treatment effects observed in the main experiment.

3.7 Heterogeneous treatment effects

We hypothesized in the pre-analysis plan that treatment responses would depend on people's pre-treatment beliefs about the labor market impact of the Mariel boatlift. Specifically, those with pessimistic pre-treatment beliefs about the Mariel boatlift should become more optimistic about the labor market impact of immigration after receiving the research evidence, and vice versa.

3.7.1 Post-treatment beliefs

As expected, we find negative and significant interaction effects of the treatment and pre-treatment beliefs about how the Mariel boatlift affected low-skilled workers on people's post-treatment beliefs about how most Americans are affected by low-skilled immigration (Panel A of Table A.16). We also observe negative interaction effects between the treatment and pre-treatment beliefs on post-treatment beliefs about the effect of immigration on people's own household, but these effects are not statistically significant.[17]

[16]Results on treatment effects for the sample of respondents who also complete the follow-experiment can be seen in Table A.10.

[17]One concern is that heterogeneous treatment effects by pre-treatment beliefs may be driven by correlations between pre-treatment beliefs and other characteristics, such as political views, race, work status, and income. We therefore decompose the total variation in pre-treatment beliefs into a component predicted by the pre-specified observables we use as control variables throughout the paper, and the residual component of pre-treatment beliefs that is not explained by these observables.

Electronic copy available at: https://ssrn.com/abstract=3017420

While we document significant treatment heterogeneity by pre-treatment beliefs about how the Mariel boatlift affected low-skilled workers, we do not find any evidence of treatment heterogeneity by pre-treatment beliefs about how the Mariel boatlift affected high-skilled workers. This could reflect the fact that the large majority of our respondents thought that the Mariel boatlift had no impact on high-skilled workers.[18]

3.7.2 Policy preferences

Our conceptual framework predicts a negative interaction effect between the treatment and pre-treatment beliefs on self-reported policy views (Section A). As predicted, we find a negative interaction effect between the treatment and pre-treatment beliefs about how the Mariel boatlift affected low-skilled workers on support for low-skilled immigration, but the estimated coefficient is only marginally significant (Panel A of Table A.17). That we find a less robust interaction effect for policy preferences than for beliefs could reflect a lack of statistical power as we observe smaller average treatment effects on policy preferences than on beliefs. This is not to say that our evidence is inconsistent with our conceptual framework. As Panel A of Figure A.6 shows, we find large and significant treatment effects conditional on having pessimistic pre-treatment beliefs, and imprecisely estimated null effects conditional on having optimistic pre-treatment beliefs. On support for high-skilled immigration, we do not observe any significant treatment heterogeneity by pre-treatment beliefs.

Reassuringly, we find very similar results using the variation in pre-treatment beliefs that is not explained by our pre-specified covariates. By contrast, the variation in pre-treatment beliefs explained by our pre-specified covariates does not predict heterogeneous responses (results available upon request). This suggests that the observed interaction effects between the treatment and pre-treatment beliefs are indeed driven by genuine variation in beliefs.

[18]These results are consistent with our respondents receiving a larger information shock about low-skilled immigrants' labor market impact compared to the labor market impact of high-skilled immigrants.

Electronic copy available at: https://ssrn.com/abstract=3017420

However, statistical power may be an issue since the average treatment effect on support for high-skilled immigration was significantly lower than on support for low-skilled immigration.

3.7.3 Heterogeneity by political affiliation and skill-level

We also pre-specified to examine heterogeneous responses on self-reported policy views by people's political affiliation and self-perceived skill levels. We find no evidence of heterogeneity based on self-identifying as Republican (Panel B of Table A.17). This result could reflect different mechanisms going in opposite directions. While Republicans have more pessimistic pre-treatment beliefs than non-Republicans—suggesting there is more scope to change their beliefs with the research evidence—they may be also be more likely to engage in partisan motivated reasoning to dismiss the research evidence. We also find no evidence of heterogeneity based on people's self-perceived skill level (Panel C of Table A.17). The lack of heterogeneity by skill level is consistent with people caring more about how immigration affects most Americans than their own household.

3.7.4 Machine-learning approaches to heterogeneity

The selection of pre-specified covariates for the heterogeneity analysis in the previous section was motivated by theoretical considerations. As a supplement to this analysis, we employ Classification and Regression Trees (CART). This machine-learning method allows us to consider all covariates in our data set and select those which predict maximum differences in the magnitude of treatment effects for different sub-populations, while simultaneously controlling for multiple hypothesis testing (Athey and Imbens, 2016). Specifically, the algorithm constructs a tree by sequentially dividing the data set into two groups with the target of minimizing the mean squared

Electronic copy available at: https://ssrn.com/abstract=3017420

error of the estimated treatment effects. To deal with over-fitting, the algorithm uses a cross-validation set to reduce the depth of the tree.[19]

Figure A.7 presents the trees obtained by the CART algorithm for the outcome variables of interest. People's pre-treatment beliefs about the wage impact of the Mariel boatlift, or their confidence in these beliefs, appear at the top of all of the trees presented in the figure. That is, the CART algorithm confirms that pre-treatment beliefs about the labor market impact of the Mariel boatlift are most powerful at predicting differences in responses to the information treatment both in terms of policy views on immigration and post-treatment beliefs about the labor market impact of immigration. Proxies for economic status (such as income and employment status) and ethnicity appear in the lower end of the trees, suggesting that economic and cultural considerations also shape people's responsiveness to the information treatment. Overall, we find it encouraging that the data-driven machine learning approach singles out the most theoretically relevant dimension—pre-treatment beliefs about the Mariel boatlift—when assessing which factors shape responses to the information.

3.8 Discussion

Our causal estimates demonstrate that labor market concerns are a quantitatively important driver of attitudes towards immigration. Do labor market concerns mostly operate through self-interested concerns about own labor market outcomes or through concerns about how immigration affects the national labor market? While our treatment changed beliefs about both dimensions, the lack of heterogeneous treatment effects by self-perceived skill levels documented in Section 3.7.3 may suggest that

[19] We specify the bin size of the final nodes to be at least 200 and use 100 cross-validations sets for the out-of-sample predictions; i.e., we randomly partition the data into equally-sized sub-samples. One of the folds is used for evaluation, while the remaining folds are utilized for fitting the tree.

Electronic copy available at: https://ssrn.com/abstract=3017420

the treatment mainly operated through concerns about how immigration affects the national labor market.

To test whether this explanation is consistent with correlational evidence, we run regressions with control group respondents simultaneously including beliefs about the labor market impact of increased immigration on "most Americans" as well on the respondents' own household. We find that beliefs about the labor market impact on "most Americans" are stronger predictors of support for immigration than beliefs about the labor market impact on people's own household (Table A.13). While a one standard deviation change in beliefs about the wage impact of immigration on most Americans is associated with a 0.12 to 0.21 standard deviation change in attitudes towards different types of immigrants, the corresponding point estimates for beliefs about the wage impact on people's own household are either zero or negative. The same pattern holds after controlling for beliefs about the cultural impact of immigration.[20] These estimates suggest an important role for nationwide labor market concerns in the formation of immigration attitudes.

This finding may explain why our main results differ from the current consensus that labor market concerns are not a quantitatively important driver of attitudes towards immigration (Hainmueller and Hopkins, 2014b). This consensus is formed by studies showing that people's policy preferences on immigration vary little with their own labor market position. While this suggests that self-interested labor market concerns are not a quantitatively important driver of attitudes towards immigration, the absence of a correlation does not rule out an important role for labor market concerns about the nationwide labor market effect of immigration.

More generally, we believe our results challenge the claim that opposition to

[20] We also replicate these correlational patterns using data from a probability-based sample of the US population (see Table A.14).

Electronic copy available at: https://ssrn.com/abstract=3017420

immigration is primarily rooted in cultural concerns. For instance, Sniderman et al. (2004) find that immigrants who are described as being unlikely to fit in culturally face much lower support than immigrants who are described as lacking professional skills. Our data on subjective beliefs illustrates the difficulty of interpreting this and similar evidence. For instance, we find that our respondents believe that immigrants *highly familiar* with American values and traditions have a 0.36 to 0.4 of a standard deviation more positive impact on the finances of most Americans than immigrants *not familiar* with American values and traditions (Table A.15). This result demonstrates that people's beliefs about the cultural impact of immigrants depend on the immigrants' skill levels. We also find that our respondents believe that high-skilled immigrants have a 0.33 of a standard deviation more positive impact on American values and traditions than low-skilled immigrants, suggesting that beliefs about the economic impact of immigrants also depend on the immigrants' cultural characteristics.

4 Concluding remarks

In this paper we present evidence that labor market concerns causally affect public support for immigration. We find that a one standard deviation change in beliefs about the labor market impact of low-skilled immigration changes attitudes towards increasing low-skilled immigration by 0.62 of a standard deviation. Signatures on real online petitions demonstrate that the research evidence affects political behavior. These findings complement a literature documenting how immigration affects electoral support for anti-immigrant parties (Dehdari, 2018; Dustmann et al., 2018; Halla et al., 2017; Mayda et al., 2018). Immigration is now said to rival economics as "the driving force in Western politics," and may continue to dominate

Electronic copy available at: https://ssrn.com/abstract=3017420

the political discussion for decades to come.[21] Understanding the extent to which opposition to immigration is due to perceived economic or cultural threats could therefore be crucial to understand the rise and persistence of ring-wing populism in the US and Europe. While natural experiments studying the impact of immigration on voting outcomes cannot identify the underlying motivations of voters, our findings suggest that labor market concerns may be an important mechanism at play.

We believe that our approach of using the results from natural experiments to measure and shock people's beliefs about the expected costs and benefits of economic policies could be applied in many different settings. A large literature has investigated whether people hold accurate beliefs about policy-relevant facts. The consensus from this literature is that people suffer from widespread political misperceptions, which undermine their ability to form meaningful opinions (Flynn et al., 2017). However, to form meaningful opinions about topics such as immigration, taxation, and monetary policy, it is arguably equally important for voters to engage in counterfactual policy analysis. We think that a promising avenue for future research is to assess the economic competence of voters by investigating two related questions: (i) which mental models people use to assess the economic implications of different policies and (ii) how people use new evidence to update their mental models of the economy.

References

Alan, Sule and Seda Ertac, "Belief in Hard Work and Prosocial Behavior: Evidence from a Randomized Field Experiment," 2017.

Alesina, Alberto, Armando Miano, and Stefanie Stantcheva, "Immigration and Redistribution," NBER Working Paper 24733 2018.

[21] Rachman, Gideon. "Migration will drive western politics for decades to come," *Financial Times,* May 8, 2018. https://www.ft.com/content/7f4c6222-4f94-11e8-9471-a083af05aea7

Electronic copy available at: https://ssrn.com/abstract=3017420

_ , **Stefanie Stantcheva, and Edoardo Teso**, "Intergenerational Mobility and Preferences for Redistribution," *American Economic Review*, 2018, *108* (2), 521–554.

Athey, Susan and Guido Imbens, "Recursive Partitioning for Heterogeneous Causal Effects," *Proceedings of the National Academy of Sciences*, 2016, *113* (27), 7353–7360.

Borjas, George J, *Immigration Economics*, Harvard University Press, 2014.

Bursztyn, Leonardo, Michael Callen, Bruno Ferman, Saad Gulzar, Ali Hasanain, and Noam Yuchtman, "Political Identity: Experimental Evidence on Anti-Americanism in Pakistan," 2016.

Card, David, "The Impact of the Mariel Boatlift on the Miami Labor Market," *Industrial and Labor Relations Review*, 1990, *43* (2), 245–257.

_ , "Is the New Immigration Really so Bad?," *Economic Journal*, 2005, *115* (507).

_ , "Comment: The Elusive Search for Negative Wage Impacts of Immigration," *Journal of the European Economic Association*, 2012, *10* (1), 211–215.

_ **and Giovanni Peri**, "*Immigration Economics* by George J. Borjas: A Review Essay," *Journal of Economic Literature*, 2016, *54* (4), 1333–1349.

_ , **Christian Dustmann, and Ian Preston**, "Immigration, Wages, and Compositional Amenities," *Journal of the European Economic Association*, 2012, *10* (1), 78–119.

Citrin, Jack, Donald P Green, Christopher Muste, and Cara Wong, "Public Opinion Toward Immigration Reform: The Role of Economic Motivations," *Journal of Politics*, 1997, *59* (3), 858–881.

Clemens, Michael A., "What the Mariel Boatlift of Cuban Refugees Can Teach Us about the Economics of Immigration: An Explainer and a Revelation," 2017.

Coffman, Lucas, Clayton R Featherstone, and Judd B Kessler, "A Model of Information Nudges," *Working paper*, 2015.

Cruces, Guillermo, Ricardo Perez-Truglia, and Martin Tetaz, "Biased Perceptions of Income Distribution and Preferences for Redistribution: Evidence from a Survey Experiment," *Journal of Public Economics*, 2013, *98*, 100–112.

de Quidt, Jonathan, Johannes Haushofer, and Christopher Roth, "Measuring and Bounding Experimenter Demand," *American Economic Review* (forthcoming), 2018.

Electronic copy available at: https://ssrn.com/abstract=3017420

Dehdari, Sirus, "Economic Distress and Support for Far-right Parties - Evidence from Sweden," available at SSRN: https://ssrn.com/abstract=3160480 2018.

DellaVigna, Stefano, John A List, Ulrike Malmendier, and Gautam Rao, "Voting to tell others," *Review of Economic Studies*, 2016, *84* (1), 143–181.

Downs, Anthony, "An Economic Theory of Political Action in a Democracy," *Journal of Political Economy*, 1957, *65* (2), 135–150.

Dustmann, Christian and Ian Preston, "Attitudes to Ethnic Minorities, Ethnic Context and Location Decisions," *Economic Journal*, 2001, *111* (470), 353–373.

_ **and** _ , "Is Immigration Good or Bad for the Economy? Analysis of Attitudinal Responses," *Research in Labor Economics*, 2006, *24* (1), 3–34.

_ **, Francesca Fabbri, and Ian Preston**, "The Impact of Immigration on the British Labour Market," *Economic Journal*, 2005, *115* (507).

_ **, Kristine Vasiljeva, and Anna Piil Damm**, "Refugee Migration and Electoral Outcomes," *Review of Economic Studies* (forthcoming), 2018.

Facchini, Giovanni, Anna Maria Mayda, and Riccardo Puglisi, "Illegal immigration and media exposure: Evidence on individual attitudes," *CEPR Discussion Papers 7593*, 2009.

_ **, Yotam Margalit, and Hiroyuki Nakata**, "Countering Public Opposition to Immigration: The Impact of Information Campaigns," 2016.

Falk, Armin and Florian Zimmermann, "A Taste for Consistency and Survey Response Behavior," *CESifo Economic Studies*, 2012, *59* (1), 181–193.

Flynn, D.J., Brendan Nyhan, and Jason Reifler, "The Nature and Origins of Misperceptions: Understanding False and Unsupported Beliefs About Politics," *Political Psychology*, 2017, *38*, 127–150.

Gerber, Alan S, Dean Karlan, and Daniel Bergan, "Does the Media Matter? A Field Experiment Measuring the Effect of Newspapers on Voting Behavior and Political Opinions," *American Economic Journal: Applied Economics*, 2009, pp. 35–52.

Gilens, Martin, "Political Ignorance and Collective Policy Preferences," *American Political Science Review*, 2001, *95* (2), 379–396.

Electronic copy available at: https://ssrn.com/abstract=3017420

Grigorieff, Alexis, Christopher Roth, and Diego Ubfal, "Does Information Change Attitudes Towards Immigrants? Representative Evidence From Survey Experiments," available at SSRN: `https://ssrn.com/abstract=2768187` 2018.

Hainmueller, Jens and Daniel J Hopkins, "The Hidden American Immigration Consensus: A Conjoint Analysis of Attitudes Toward Immigrants," *American Journal of Political Science*, 2014, *59* (3), 529–548.

_ **and** _ , "Public Attitudes Toward Immigration," *Annual Review of Political Science*, 2014, *17*, 225–249.

_ **and Michael J Hiscox**, "Attitudes Toward Highly Skilled and Low-skilled Immigration: Evidence from a Survey Experiment," *American Political Science Review*, 2010, *104* (01), 61–84.

_ **,** _ **, and Yotam Margalit**, "Do Concerns about Labor Market Competition Shape Attitudes Toward Immigration? New Evidence," *Journal of International Economics*, 2015, *97* (1), 193–207.

Halla, Martin, Alexander F Wagner, and Josef Zweimüller, "Immigration and Voting for the Far Right," *Journal of the European Economic Association*, 2017, *15* (6), 1341–1385.

Hopkins, Daniel J, John Sides, and Jack Citrin, "The Muted Consequences of Correct Information About Immigration," *Journal of Politics* (forthcoming), 2018.

Iyengar, Shanto, Simon Jackman, Solomon Messing, Nicholas Valentino, Toril Aalberg, Raymond Duch, Kyu S Hahn, Stuart Soroka, Allison Harell, and Tetsuro Kobayashi, "Do attitudes about immigration predict willingness to admit individual immigrants? A cross-national test of the person-positivity bias," *Public Opinion Quarterly*, 2013, *77* (3), 641–665.

Karadja, Mounir, Johanna Mollerstrom, and David Seim, "Richer (and Holier) Than Thou? The Effect of Relative Income Improvements on Demand for Redistribution," *Review of Economics and Statistics*, 2017, *99* (2), 201–212.

Kruger, Justin and David Dunning, "Unskilled and unaware of it: how difficulties in recognizing one's own incompetence lead to inflated self-assessments.," *Journal of Personality and Social Psychology*, 1999, *77* (6), 1121.

Kuklinski, James H, Paul J Quirk, Jennifer Jerit, David Schwieder, and Robert F Rich, "Misinformation and the Currency of Democratic Citizenship," *Journal of Politics*, 2000, *62* (3), 790–816.

Electronic copy available at: https://ssrn.com/abstract=3017420

Kuziemko, Ilyana, Michael I Norton, Emmanuel Saez, and Stefanie Stantcheva, "How Elastic are Preferences for Redistribution? Evidence from Randomized Survey Experiments," *American Economic Review*, 2015, *105* (4), 1478–1508.

Lergetporer, Philipp, Marc Piopiunik, Lisa Simon et al., "Do Natives' Beliefs About Refugees' Education Level Affect Attitudes Toward Refugees? Evidence from Randomized Survey Experiments," Technical Report, CESifo Group Munich 2017.

Mayda, Anna Maria, "Who is Against Immigration? A Cross-country Investigation of Individual Attitudes Toward Immigrants," *Review of Economics and Statistics*, 2006, *88* (3), 510–530.

_ , Giovanni Peri, and Walter Steingress, "The Political Impact of Immigration: Evidence from the United States," 2018.

Mummolo, Jonathan and Erik Peterson, "Demand Effects in Survey Experiments: An Empirical Assessment," available at SSRN: `https://ssrn.com/abstract=2956147` 2018.

Scheve, Kenneth F and Matthew J Slaughter, "Labor Market Competition and Individual Preferences over Immigration Policy," *Review of Economics and Statistics*, 2001, *83* (1), 133–145.

Sides, John and Jack Citrin, "How Large the Huddled Masses? The Causes and Consequences of Public Misperceptions About Immigrant Populations," in "Annual Meeting of the Midwest Political Science Association, Chicago" 2007.

Smith, Tom W., Michael Davern, Jeremy Freese, and Michael Hout, "GSS General Social Survey," Data accessed January 30, 2018 from the GSS Data Explorer website at `gssdataexplorer.norc.org` 2014.

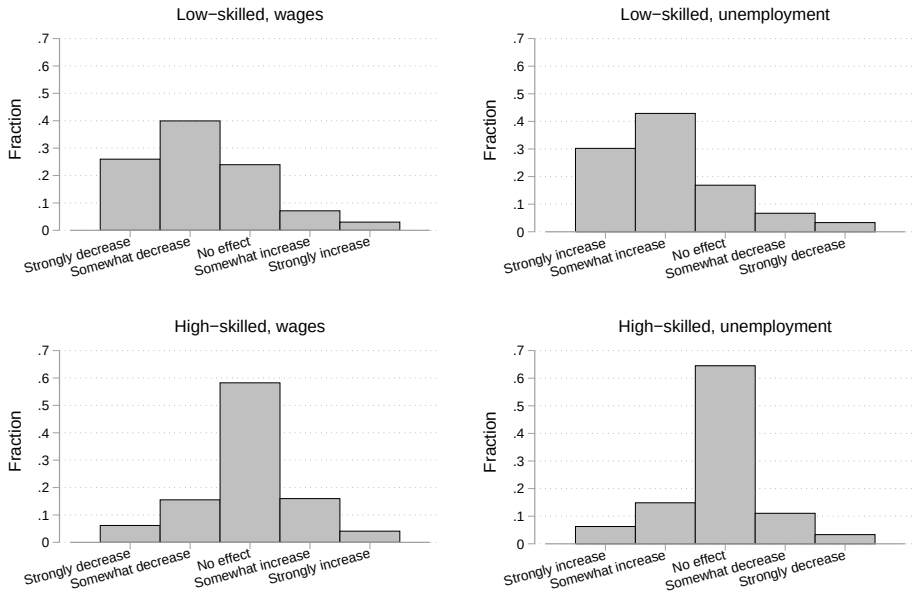
Sniderman, Paul M, Louk Hagendoorn, and Markus Prior, "Predisposing Factors and Situational Triggers: Exclusionary Reactions to Immigrant Minorities," *American Political Science Review*, 2004, *98* (01), 35–49.

Zizzo, Daniel John, "Experimenter Demand Effects in Economic Experiments," *Experimental Economics*, 2010, *13* (1), 75–98.

Electronic copy available at: https://ssrn.com/abstract=3017420

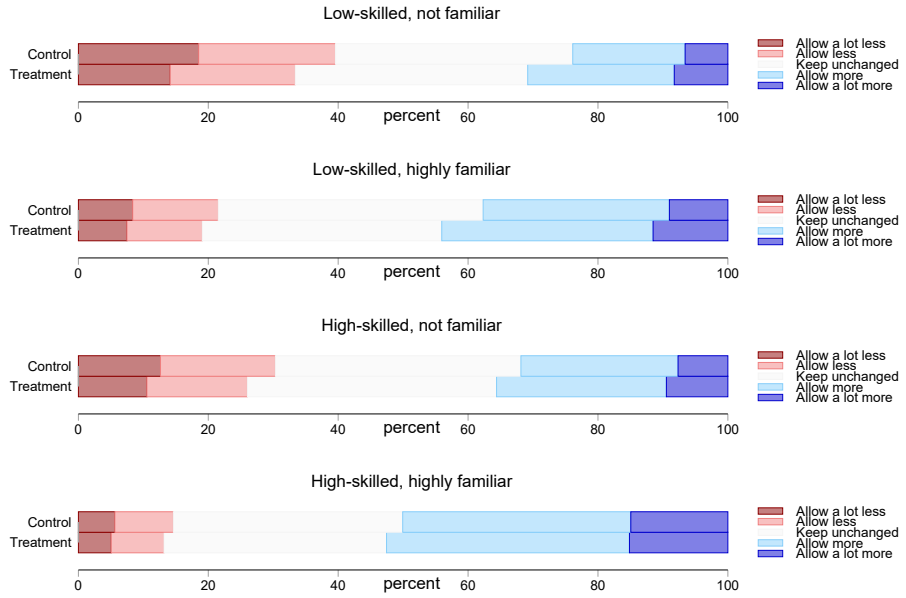
Main figures

Figure 1: Pre-treatment beliefs about the labor market impacts of the Mariel boatlift



Notes: The figure shows the distribution of pre-treatment beliefs about the impact of the Mariel boatlift on wages (left panels) and unemployment (right panels) for low-skilled workers (top panels) and high-skilled workers (bottom panels). Respondents are asked two question on wages: "In the five-year period after 1980, how do you think wages of low-skilled (high-skilled) workers in Miami were affected by the mass immigration of Cubans?" The responses are on a five-point scale ranging from (1) Strongly decrease to (5) Strongly increase. They are also asked two questions about unemployment: "In the five-year period after 1980, how do you think unemployment among low-skilled (high-skilled) workers in Miami was affected by the mass immigration of Cubans?" The responses are on a five-point scale ranging from (1) Strongly increase to (5) Strongly decrease.

Figure 2: Distribution of attitudes towards immigrants: Treatment vs. control



Notes: The figure shows the distribution of attitudes towards low-skilled/high-skilled immigrants that are highly familiar/not familiar with American values and traditions, disaggregated by the treatment and control group.

Electronic copy available at: https://ssrn.com/abstract=3017420

Main tables

Table 1: Beliefs about the labor market impact of immigration (post-treatment)

	Low-skilled			High-skilled			Follow-up
	Wages	Employment	Index	Wages	Employment	Index	Overall Labor Concerns
Panel A: With controls							
Treatment	0.163	0.156	0.159	0.197	0.264	0.230	0.117
	(0.046)	(0.047)	(0.042)	(0.048)	(0.048)	(0.043)	(0.040)
Observations	1474	1469	1469	1476	1470	1470	2075
Panel B: Without controls							
Treatment	0.169	0.164	0.166	0.194	0.259	0.228	0.139
	(0.048)	(0.049)	(0.044)	(0.049)	(0.049)	(0.044)	(0.043)
Observations	1474	1469	1469	1476	1470	1470	2075

Notes: The table shows OLS regression results where the dependent variables are beliefs about the labor market impact of immigration. In the main study, respondents were asked how they thought admitting more low-skilled/high-skilled immigrants would affect wages and unemployment for "most Americans." We randomized whether respondents answered these questions for low-skilled or high-skilled immigrants. In the follow-up, respondents were asked whether they thought increased immigration would hurt American workers. All questions were answered on 5-point Likert scales where higher values indicate more optimistic views on the effect of immigration. The outcome variables are z-scored using the mean and standard deviation in the control group. We include controls as described in the pre-analysis plan (also listed in Table 2). Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table 2: Self-reported attitudes towards immigration (post-treatment)

	Low-skilled (Main Study)			High-skilled (Main Study)			Follow-up	
	Not familiar	Familiar	Index	Not familiar	Familiar	Index	Low-skilled	High-skilled
Panel A: With controls								
Treatment	0.170	0.112	0.141	0.105	0.040	0.072	0.096	0.064
	(0.033)	(0.034)	(0.031)	(0.033)	(0.034)	(0.031)	(0.040)	(0.042)
Adjusted p-value	[0.001]	[0.001]		[0.003]	[0.135]			
Observations	3130	3130	3130	3130	3130	3130	2075	2075
Panel B: Without controls								
Treatment	0.168	0.117	0.143	0.108	0.046	0.077	0.108	0.070
	(0.036)	(0.036)	(0.033)	(0.036)	(0.035)	(0.032)	(0.044)	(0.043)
Adjusted p-value	[0.001]	[0.001]		[0.005]	[0.108]			
Observations	3130	3130	3130	3130	3130	3130	2075	2075

Notes: The table shows OLS regression results where the dependent variables are attitudes to the different types of immigrants. The answers were given on a five point scale from 1: "Allow a lot less of these immigrants" to 5: "Allow a lot more of these immigrants." The question order was randomized (statistical tests show no order effects). The outcomes are z-scored using the mean and standard deviation in the control group. The indices are created by taking the mean of the responses to immigrants with different familiarity with American values and traditions for each skill level. Adjusted p-values are in brackets. "Treatment" is an indicator equal to 1 if respondents received the research evidence. Controls include gender, age, ethnicity, region, household size, household income, education, employment status, party affiliation, whether the respondent was born in the US, whether the subject's parents were born in the US, self-perceived skill-level, and pre-treatment beliefs about the labor market impact of low-skilled (or high-skilled) immigration and are coded as described in the pre-analysis plan. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table 3: Online petitions on H-2B visas (post-treatment)

	Intention: H2B Visas			Actual signatures H2B Visas	
	Increase	Decrease	Net support	Increase	Decrease
Panel A: With controls					
Treatment	0.048	-0.058	0.135	–	–
	(0.016)	(0.016)	(0.034)	–	–
Observations	3130	3130	3130	–	–
Panel B: Without controls					
Treatment	0.048	-0.061	0.141	0.027	-0.004
	(0.017)	(0.016)	(0.036)	(0.007)	(0.007)
Control Mean	0.286	0.322	-0.000	0.039	0.045
Observations	3130	3130	3130	3130	3130

Notes: The three first columns show regression results where the dependent variable is intention to sign the petitions. "Increase" ("Decrease") is an indicator equal to 1 if a respondent wanted to sign the petition suggesting to increase (decrease) the annual cap on the H-2B visa program. "Net support" is a z-scored transformation of a variable taking value 1 (-1) if a respondent wanted to sign the petition to increase (decrease) the annual cap on the H-2B visa program and 0 otherwise. The two last columns show actual signatures. Since we only observe actual signatures on the treatment group level, we cannot include controls and run regressions for these outcomes. To do testing, we calculate standard errors using the standard formula for proportion tests. "Treatment" is an indicator equal to 1 if respondents received information about the labor market impact of the Mariel boatlift. We use the same controls as in Table 2. For the questions on intention to sign the petitions, we apply robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table 4: The causal effect of labor market concerns on support for low-skilled immigrants: Subsample of respondents with pessimistic pre-treatment beliefs

	Labor Market Concerns	Support	Support	
	OLS (reduced form)	OLS (reduced form)	IV	OLS
Panel A: Controls				
Treatment	0.222	0.148		
	(0.043)	(0.033)		
Labor Market Concerns			0.622	0.318
			(0.212)	(0.042)
Observations	1218	2599	1218	624
Panel B: Additional controls				
Treatment	0.185	0.125		
	(0.033)	(0.044)		
Labor Market Concerns			0.677	0.282
			(0.249)	(0.052)
Observations	1203	1203	1203	613

Notes: These regressions are based on the subsample of respondents who thought that the Mariel boatlift had either negative or no effects on wages and unemployment. Column 4 only uses control group respondents, while Colums 1 to 1 use both treatment and control group respondents. The dependent variable in columns 2 to 4 is an index for attitudes towards low-skilled immigrants. The dependent variable in column 1 is "Labor Market Concerns" which is an unweighted index of people's beliefs about the effect of low-skilled immigration on (i) "wages for most Americans" and (ii) "job opportunities or job security for most Americans." In Column 3 we instrument "Labor Market Concerns" with the treatment indicator. In Columns 1, 2, and 4, we run OLS regressions. We randomized whether people were asked about the impact of low-skilled or high-skilled immigrants. All variables have been standardized. All regressions include the controls described in Table 2. Panel B additionally controls for people's beliefs about the impact of immigration on wages and employment of the own household, as well as beliefs about the tax burden created by immigrants for most Americans and the own household, as well as beliefs about the cultural impact of immigration. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

For online publication only:

Labor Market Concerns and Support for Immigration

Ingar Haaland and Christopher Roth

Summary of the Online Appendix

Section A provides a conceptual framework. Section B lists some minor deviations from the pre-analysis plans. Section C.1 provides additional figures. Figure A.1 shows an overview of the structure of the experiment. Figure A.2 displays the correlates of demographics with people's pre-treatment beliefs about the wage and employment impact of the Mariel boatlift. Figure A.3 displays the distribution of beliefs about the labor market impact of immigration on most Americans versus own household. Figure A.4 shows treatment effects in absolute and relative changes. Figure A.5 shows heterogeneous updating of beliefs about the labor market effects of immigrants. Figure A.6 shows heterogeneous treatment effects on support for low-skilled immigration by people's pre-treatment beliefs about the Mariel boatlift. Figure A.7 shows the causal trees obtained by the CART algorithm. Figure A.8 displays the distribution of preferences and beliefs using control group respondents.

Section C.2 provides evidence on covariate balance and describes the sample. Tables A.1 and A.2 provide descriptive statistics. Table A.3 shows covariate balance for the main sample, while Table A.4 provides evidence on covariate balance for the sample in the follow-up study. Table A.5 examines correlates of attrition in the follow-up study.

Section C.3 provides descriptives and further evidence on the mechanisms.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.6 summarizes treatment effects on people's beliefs about how their own household's and most American's wages, employment, tax burden and American culture are affected by low-skilled immigration. Table A.7 shows treatment effects on people's beliefs about the overall economic impact. Table A.8 shows the IV estimates on people's support for low-skilled immigration. Table A.9 shows the main outcome questions for people with pessimistic pre-treatment beliefs. Table A.10 shows the main results on policy preferences restricting the sample to those who also completed the follow-up survey.

Table A.11 describes correlates of people's beliefs about the labor market impact of the Mariel boatlift. Table A.12 shows correlates of people's attitudes. Tables A.13 and A.14 examine the relative importance of self-interested vs. group-level concerns in shaping immigration preferences. Table A.15 describes how people's beliefs about the economics and cultural impact of immigrants covary with economic and cultural characteristics of immigrants.

Section C.4 examines heterogeneity in treatment effects. Table A.16 explores heterogeneous treatment effects on people's beliefs about the labor market impact of immigration. Table A.17 and Table A.18 explore heterogeneity for our main self-reported outcome questions by people's self-perceived skill level, their political affiliation, and their pre-treatment belief about the effect of the Mariel boatlift. Table A.19 examines heterogeneous treatment effects along these dimensions on people's intention to sign the petition.

In Section C.5, Figure A.9 describes our respondents' beliefs about the survey and beliefs about the relevance, accuracy, and trustworthiness of the research evidence. Table A.20 explores how beliefs about the research study are correlated with support for immigration. Table A.21 examines how standard demographics are correlated with beliefs about the research study.

Electronic copy available at: https://ssrn.com/abstract=3017420

Section D provides screenshots of the online petition, the invitation email for the follow-up study, and the consent forms in the main study and the obfuscated follow-up study. Finally, Section E and F provide experimental instructions for, respectively, the main experiment and the obfuscated follow-up.

A Conceptual framework

We here present a simple theoretical framework, adapted from Card et al. (2012), on how people report their policy views on immigration. Individual i's indirect utility function depends on her net income and the characteristics of her community:

$$u_i(w_i - t_i, \mathbf{c}_i)$$

where w_i is i's gross income; t_i is taxes; and $\mathbf{c}_i$ is a vector of characteristics of i's community, such as ethnic composition and religious diversity.[1] In the paper, we refer to net income, $w_i - t_i$, and community characteristics, $\mathbf{c}_i$, as respectively "economic" and "cultural" factors. When asked about their self-reported policy preferences on immigration, individuals compare a hypothetical scenario with more immigrants (w_i',t_i',$\mathbf{c}_i'$) to the current situation (w_i°,t_i°,$\mathbf{c}_i^\circ$). The individuals then report a transformation of the difference in the indirect utilities:

$$y_i = g_i[(w_i', t_i', \mathbf{c}_i') - (w_i^\circ, t_i^\circ, \mathbf{c}_i^\circ)]$$

[1] Card et al. (2012) use the term "compositional amenities" for these factors.

Electronic copy available at: https://ssrn.com/abstract=3017420

where g_i is an individual-specific (and increasing) response function. Taking a first-order approximation, the observed response of individual i is

$$y_i = g_i' \times \frac{\partial u_i}{\partial w_i}(\Delta w_i - \Delta t_i) + g_i' \times \frac{\partial u_i}{\partial a_i}\Delta a_i + \varepsilon_i \tag{1}$$

where $\Delta w_i = w_i' - w_i^\circ$ is the difference in perceived gross earnings between the alternative scenarios (with analogous definitions for Δt_i and $\Delta \mathbf{c}_i$) and ε_i is an idiosyncratic error term. In this model, we can define labor market concerns as $g_i' \times \frac{\partial u_i}{\partial w_i}\Delta w_i$.

To identify the role of labor market concerns in driving support for immigration without making any restrictive assumptions, one needs an instrument that exogenously shifts people's perceived labor market impact of immigration, Δw. We use a randomized information treatment to achieve this. Our treatment informs people that immigration has virtually no impact on the labor market; i.e., that $\Delta w \approx 0$. The model thus predicts that the treatment should increase support for immigration if people on average expect negative labor market impacts of immigration and treated respondents (at least partly) trust the information.

The treatment needs to satisfy two conditions to be a valid instrument for labor market concerns. First, the treatment should change beliefs about the labor market impact immigration, Δw_i, but not simultaneously change beliefs about the impact of increased immigration on taxes, Δt_i, or culture, $\Delta \mathbf{c}_i$. We ask post-treatment questions about the expected impact of immigration on both taxes and culture to test these assumptions. Second, the treatment should not affect the reporting function, g_i. While this assumption is difficult to test, we argue that concerns about temporary changes in the reporting function are mitigated through the obfuscated follow-up.

Electronic copy available at: https://ssrn.com/abstract=3017420

B Deviations from the pre-analysis plan

We posted the original pre-analysis plan on May 31, 2017 before starting the data collection. We uploaded an updated pre-analysis plan June 7, before starting collecting the data collection for the obfuscated follow-up, that was identical to the original pre-analysis plan in all aspects except for some minor changes in the instructions for the obfuscated follow-up study. The reader should consult the updated pre-analysis plan to get the actual instructions used in the experiment (also included in Section F). Some minor deviations from the pre-analysis plan are listed below:

- We pre-specified a sample size of 3000 respondents. The survey provider delivered a sample of 3130 respondents who completed the main outcome variables of interest. Out of those, 2883 respondents completed all questions in the main study.

- Because 247 respondents did not complete all the demographic questions asked at the end of the main survey, we have some missing observations for these questions. For all questions with missing values, we included a dummy for missing observations to account for this. We also supplemented some missing values from the background questions in the obfuscated follow-up survey.

- We specified applying a proportion test to the difference in the proportion of signatures on the petition in favor of increasing the annual cap and the petition in favor of decreasing the annual cap. We realized that the proportion test can only be applied to binary variables, so we applied it for each of the two petitions separately.

- As discussed in Section 3, we did not pre-specify the results presented in

Electronic copy available at: https://ssrn.com/abstract=3017420

Section 3.2 and Section 3.7.1 on belief updating; the IV results in Section 3.5; and the machine-learning approaches to heterogeneous treatment effects in Section 3.7.4.

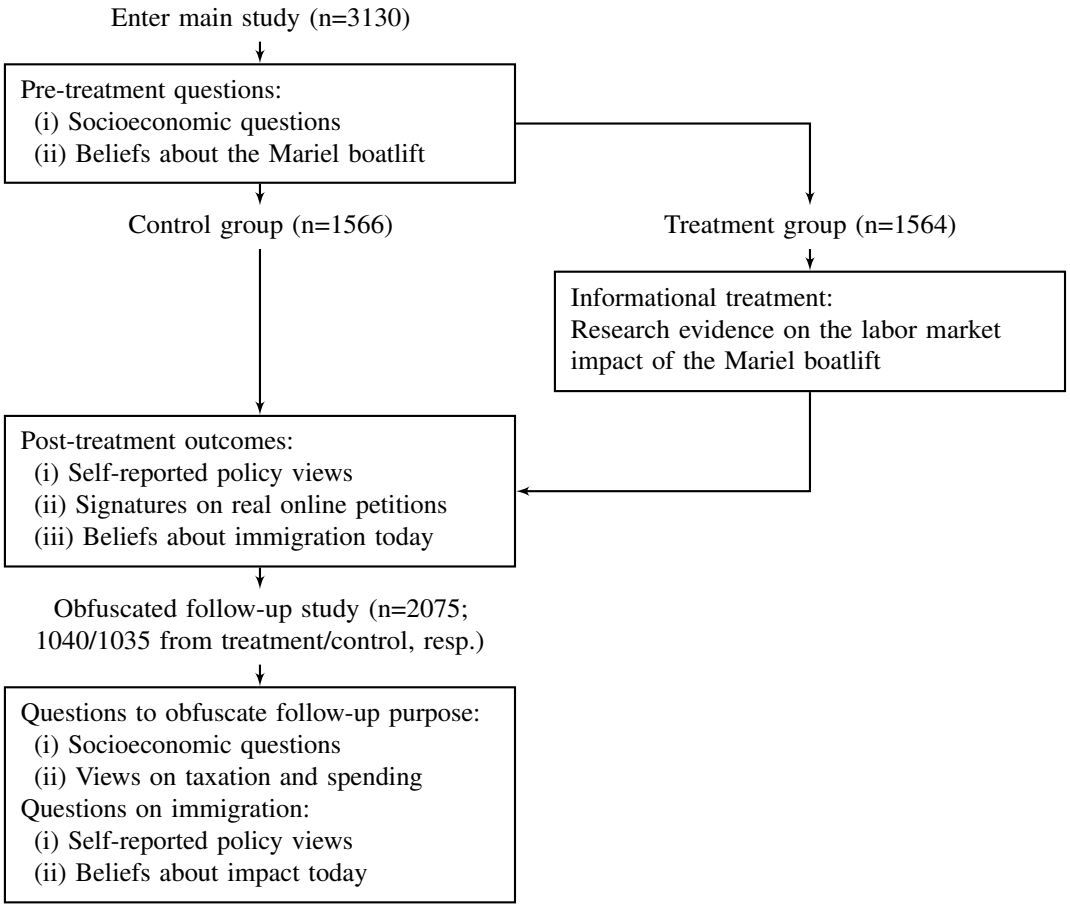
- When analyzing heterogeneous treatment effects on self-reported attitudes by pre-treatment beliefs, we pre-specified to only include the pre-treatment beliefs about the effect of the Mariel boatlift on low-skilled (high-skilled) workers when analyzing support for low-skilled (high-skilled) immigration. We later concluded that it was more informative to include interaction terms for both beliefs in both specifications. The pre-specified table is still included in the appendix (Table A.18).

Electronic copy available at: https://ssrn.com/abstract=3017420

C Figures

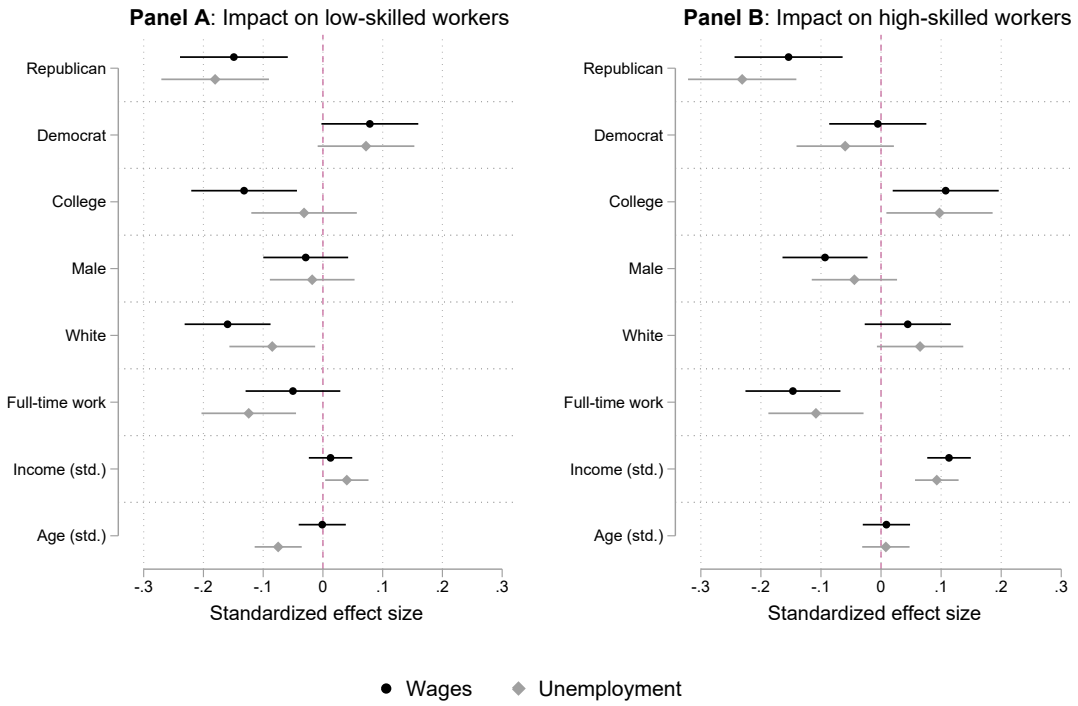
C.1 Additional figures

Figure A.1: Overview of the experiment



Electronic copy available at: https://ssrn.com/abstract=3017420

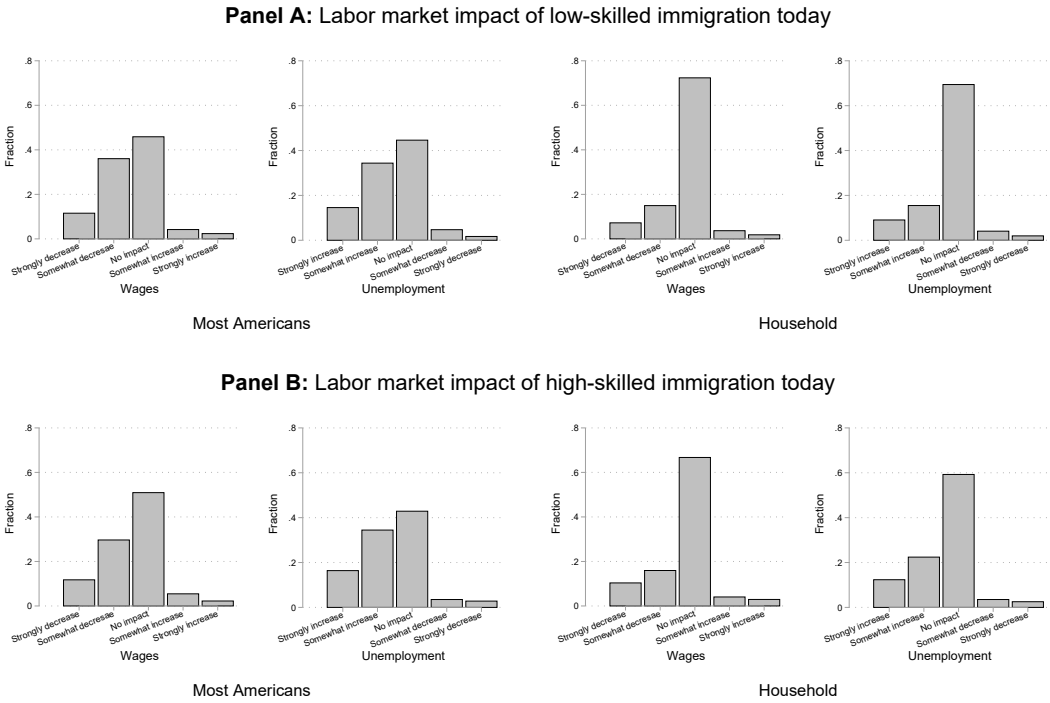
Figure A.2: Correlates of pre-treatment beliefs about the Mariel boatlift



Notes: The dots indicate the mean values of the estimated multiple regression coefficients. The lines indicate 95 percent confidence interval of the mean. In **Panel A**, the outcome variables are people's beliefs about the labor market impact of the Mariel boatlift on low-skilled workers; in **Panel B**, the outcome variables are beliefs about the labor market impact on high-skilled workers.

Electronic copy available at: https://ssrn.com/abstract=3017420

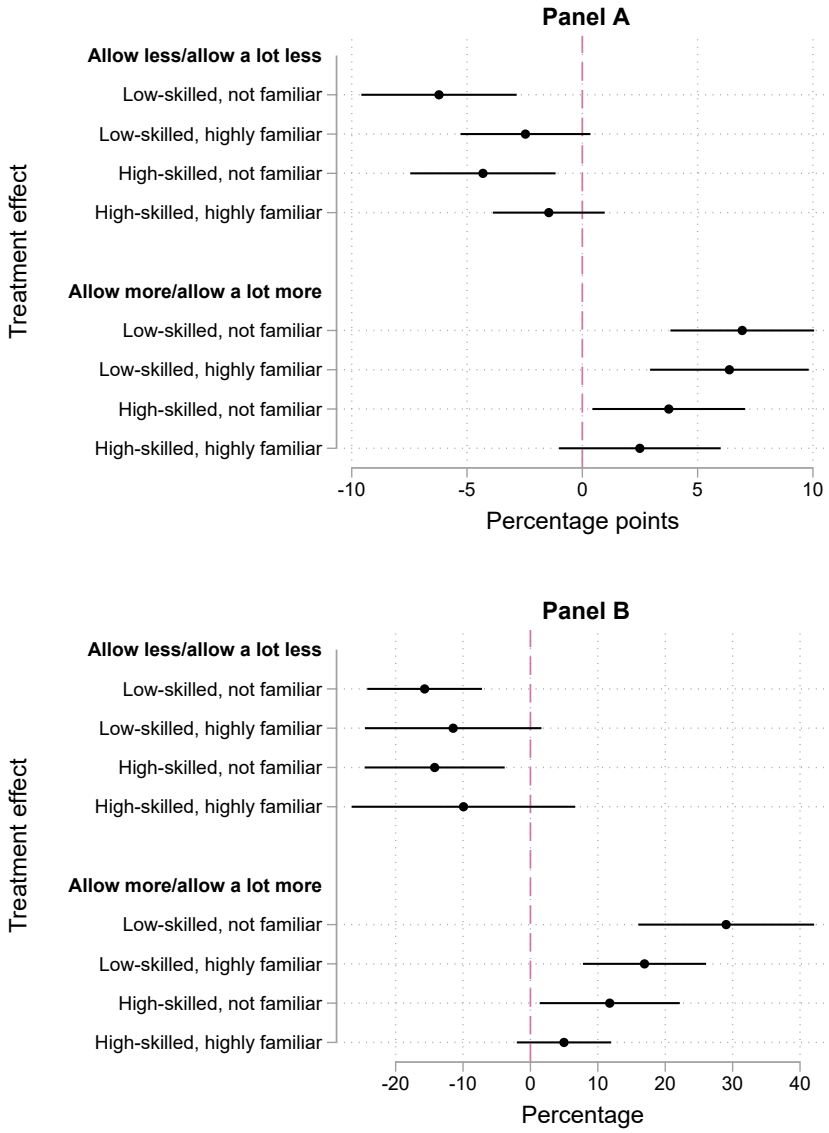
Figure A.3: Beliefs about the labor market impact of immigration: Impact on "most Americans" versus own household



Notes: **Panel A** shows the distribution of beliefs about the labor market impact of low-skilled immigrants on *most Americans* as well as on the the respondents' own household. **Panel B** show the corresponding distributions for beliefs about high-skilled immigrants. Both panels only include responses from respondents in the control group.

Electronic copy available at: https://ssrn.com/abstract=3017420

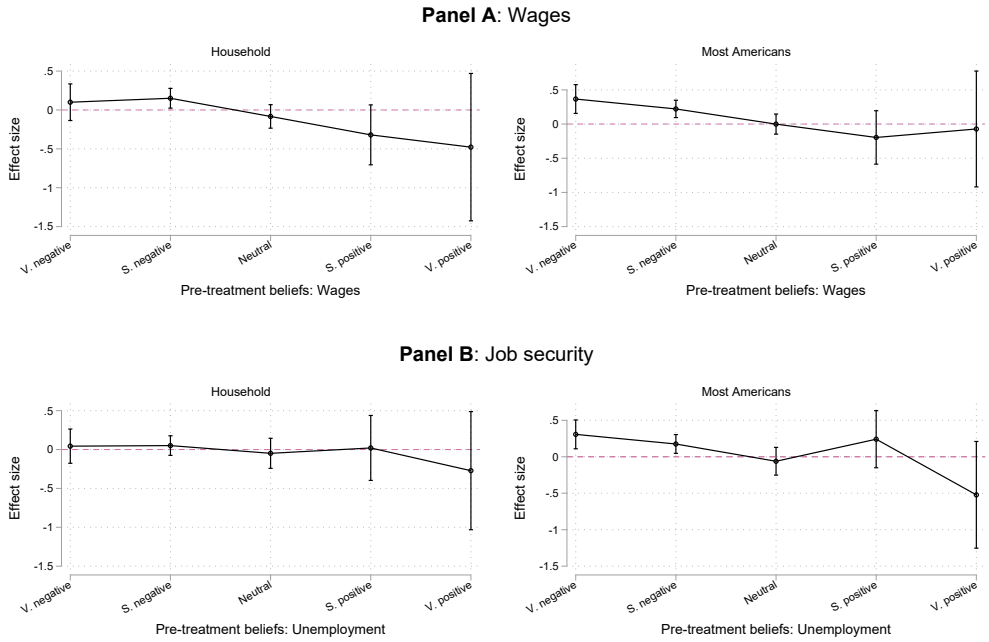
Figure A.4: Treatment effects on attitudes towards immigrants



Notes: The figure shows treatment effects on the fraction of respondents who answer that they want to allow more/allow a lot more and allow less/a lot less of low-skilled/high-skilled immigrants that are highly familiar/not familiar with American values and traditions. **Panel A** shows the treatment effects in percentage points changes. **Panel B** shows the treatment effects in percent changes. Lines indicate 95 percent confidence intervals.

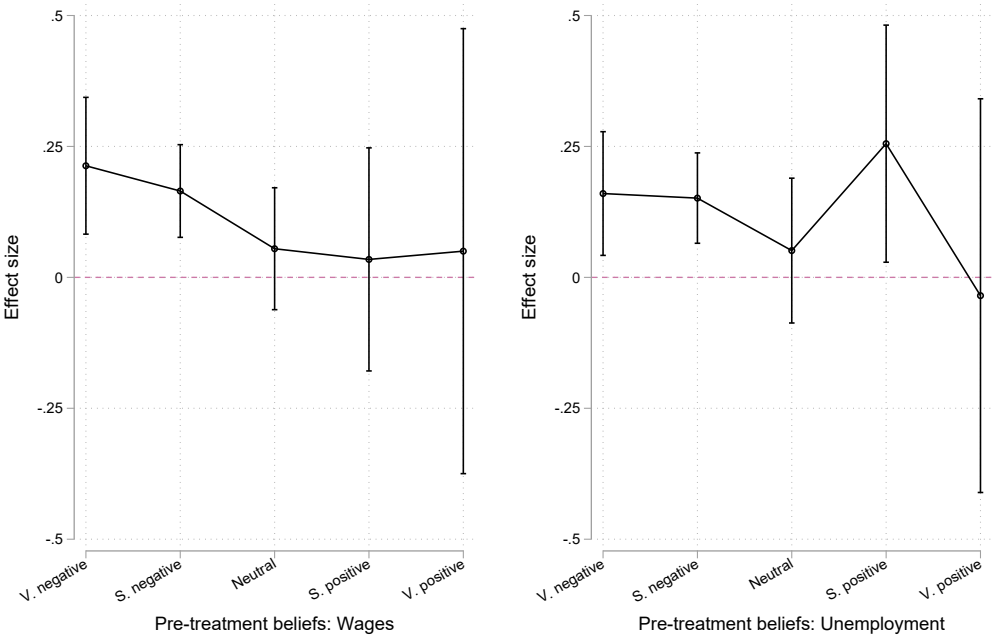
Electronic copy available at: https://ssrn.com/abstract=3017420

Figure A.5: Heterogeneity in belief updating: Labor market impact of low-skilled immigration



Notes: The figure shows standardized treatment effects on beliefs about the wage and unemployment impact of low-skilled immigration by their pre-treatment beliefs about the wage and unemployment impact of the Mariel boatlift. The outcomes are based on the follow questions: how do you think admitting more low-skilled immigrants would affect (i) "wages," and (ii) "job opportunities and job security" for their own household as well as for most Americans. The regressions include pre-specified controls (listed in Table 2). 95 percent confidence intervals are indicated.

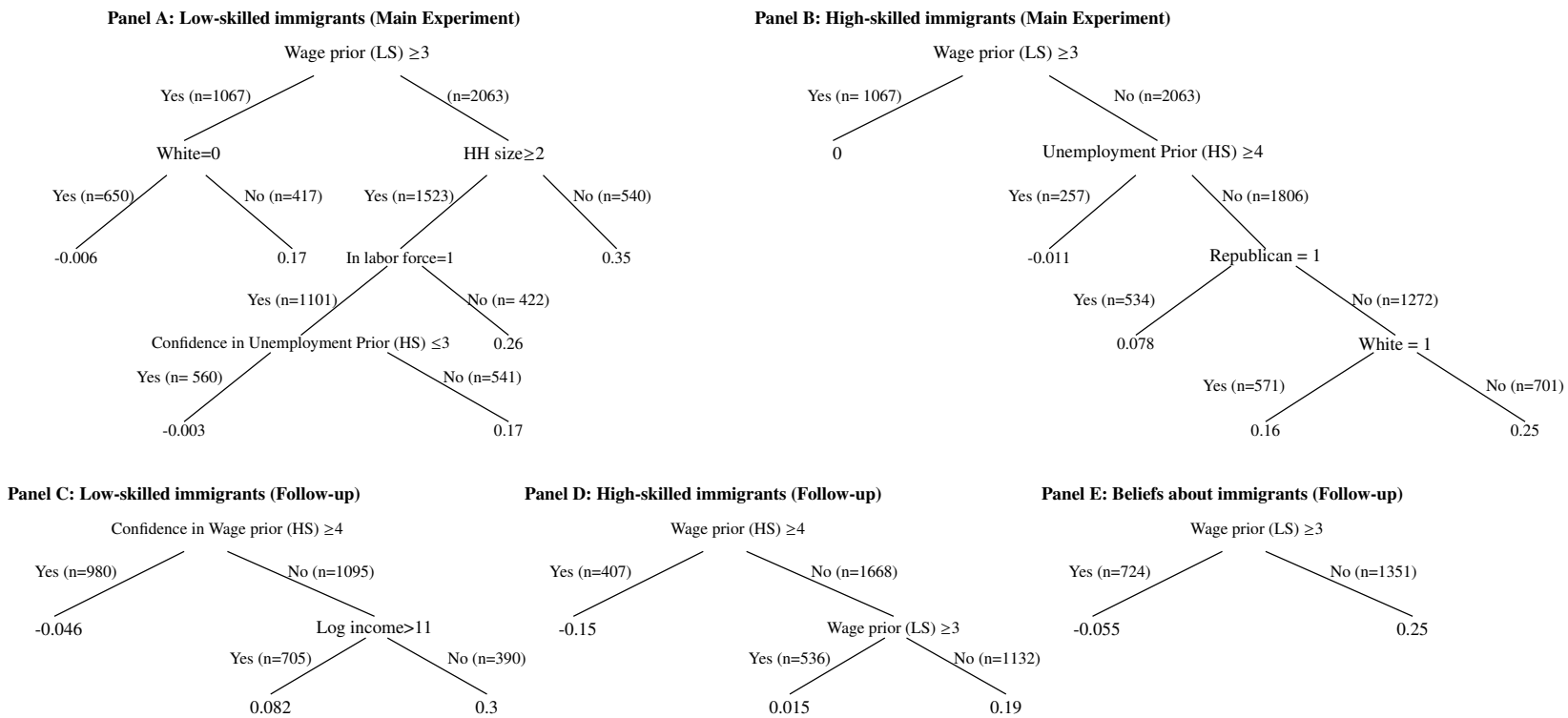
Figure A.6: Support for low-skilled immigration, treatment heterogeneity by pre-treatment beliefs



Notes: The figure shows heterogeneous treatment effects on support for low-skilled immigration based on people's pre-treatment beliefs about the low-skilled wage and unemployment impact of the Mariel boatlift. The regressions include pre-specified controls (listed in Table 2). 95 percent confidence intervals are indicated.

Electronic copy available at: https://ssrn.com/abstract=3017420

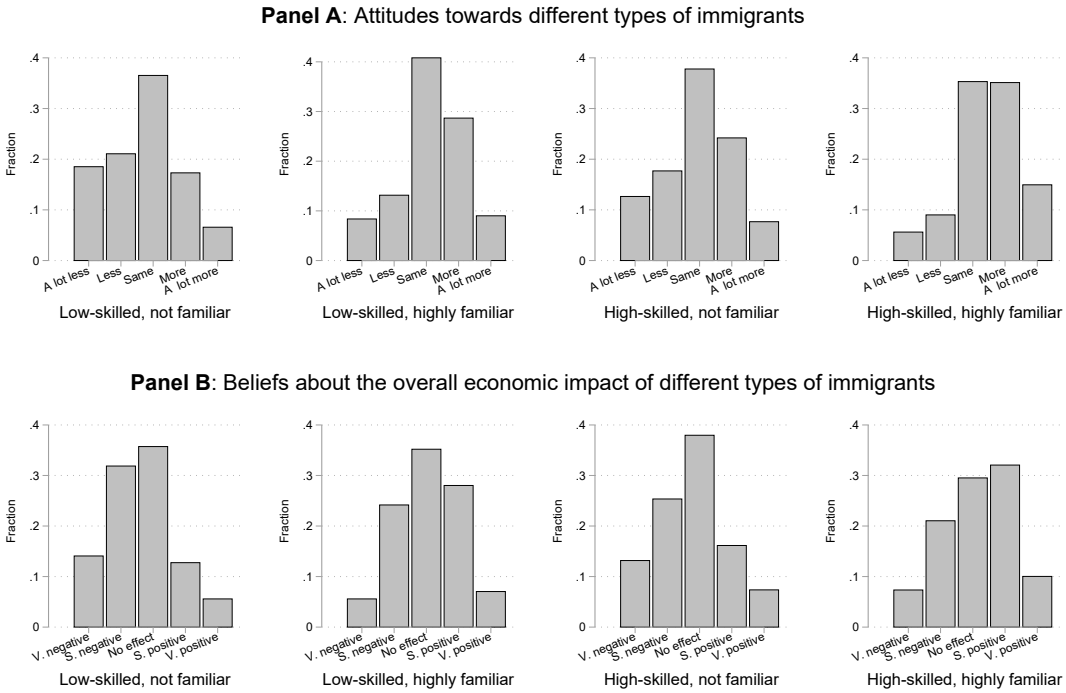
Figure A.7: Heterogeneous treatment effects: Machine Learning Classification and Regression Trees



Notes: This figure displays causal trees using Classification and Regression Trees (CART; Athey and Imbens, 2016). "Wage prior (LS)" measures people's beliefs about the impact of the Mariel boatlift on wages of low-skilled workers in Miami. "Wage prior (HS)" measures people's beliefs about the impact of the Mariel boatlift on wages of high-skilled workers in Miami. The responses for these variables are on a five-point scale ranging from (1) Strongly decrease to (5) Strongly increase. "Unemployment prior (HS)" measures people's beliefs about the impact of the Mariel boatlift on unemployment of high-skilled workers in Miami. The responses for these variables are on a five-point scale ranging from (1) Strongly increase to (5) Strongly decrease. "Confidence in Wage prior (HS)" measures people's confidence in their prior beliefs about the wage effects of the Mariel boatlift on high-skilled workers. Responses are measured on a five-point scale ranging from (1) Very unsure to (5) Very sure. "Confidence in Unemployment prior (HS)" measures people's confidence in their prior beliefs about the employment effects of the Mariel boatlift. Responses are measured on a five-point scale ranging from (1) Very unsure to (5) Very sure. White takes value one for respondents identifying as Whites, and zero otherwise. "HH size" denotes people's household size. "In labor force" takes value 1 for respondents who self-report to be in the labor force. Log income denotes the log of respondents' self-reported household income in US dollars (log income of 11 corresponds to an annual household income of approximately 60,000 USD). The numbers at the final nodes denote the z-scored treatment effects for the different subgroups defined by the causal tree. We include all variables displayed in the balance Table A.3 in the machine learning algorithm, and we set the minimum number of the final node as 200 and the number of cross validation sets as 100.

Electronic copy available at: https://ssrn.com/abstract=3017420

Figure A.8: Heterogeneity in policy views and beliefs about immigration



Notes: **Panel A** shows the distribution of policy views on whether to increase or decrease the number of low-skilled/high-skilled immigrants that are highly familiar/not familiar with American values and traditions. **Panel B** shows the distribution of beliefs about the overall economic impact on "most Americans" for the different types of immigrants. Both panels only include responses from respondents in the control group.

Electronic copy available at: https://ssrn.com/abstract=3017420

C.2 Summary statistics, balance and attrition

Table A.1: Summary statistics

	Mean	SD	Median	Min.	Max.	Obs.
Gender	0.48	0.50	0.00	0.00	1.00	3130
What is your age?	46.64	15.59	49.50	21.00	69.50	3130
Non-Hispanic White	0.46	0.50	0.00	0.00	1.00	3130
African American/Black	0.06	0.23	0.00	0.00	1.00	3130
Republican	0.25	0.43	0.00	0.00	1.00	3130
Democrat	0.37	0.48	0.00	0.00	1.00	3130
Independent	0.36	0.48	0.00	0.00	1.00	3130
Northeast	0.22	0.41	0.00	0.00	1.00	3130
Midwest	0.18	0.39	0.00	0.00	1.00	3130
West	0.24	0.43	0.00	0.00	1.00	3130
Household size	2.36	1.49	2.00	0.00	10.00	3130
Total household income	72871.41	50712.13	62500.00	7500.00	225000.00	3130
Education	4.89	2.11	5.00	0.00	9.00	3130
High-skilled	0.75	0.43	1.00	0.00	1.00	3130
Full-time employee	0.45	0.50	0.00	0.00	1.00	3130
Part-time employee	0.08	0.27	0.00	0.00	1.00	3130
Self-employed or small business owner	0.06	0.23	0.00	0.00	1.00	3130
Unemployed and looking for work	0.05	0.21	0.00	0.00	1.00	3130
Student	0.03	0.16	0.00	0.00	1.00	3130
Not in labor force	0.26	0.44	0.00	0.00	1.00	3130
College	0.78	0.42	1.00	0.00	1.00	3130
Beliefs about effect of immigration on wages of low-skilled workers	2.21	1.00	2.00	1.00	5.00	3130
Beliefs about effect of immigration on wages of high-skilled workers	2.96	0.85	3.00	1.00	5.00	3130
Beliefs about effect of immigration on unemployment of low-skilled workers	2.10	1.01	2.00	1.00	5.00	3130
Beliefs about effect of immigration on unemployment of high-skilled workers	2.90	0.80	3.00	1.00	5.00	3130
Confidence in beliefs about wages of low-skilled workers	3.49	0.97	3.00	1.00	5.00	3130
Confidence in beliefs about unemployment of low-skilled workers	3.55	0.97	4.00	1.00	5.00	3130
Confidence in beliefsabout wages of high-skilled workers	3.52	0.97	3.00	1.00	5.00	3130
Confidence in beliefs about unemployment of high-skilled workers	3.50	0.96	3.00	1.00	5.00	3130

Notes: This table displays the summary statistics for our sample.

Table A.2: Characteristics of our sample compared to the US Census

	Mean: Online sample	Mean: Online sample – follow-up	Mean: ACS
Male	0.48	0.50	0.51
What is your age?	46.6	47.4	47.1
Northeast	0.22	0.23	0.179
Midwest	0.18	0.19	0.211
West	0.24	0.23	0.24
Total household income	72871	73499	82433

Notes: This table summarizes the characteristics of our sample in the main survey as well as the follow-up survey along targeted dimensions as well as the characteristics of the 2015 American Community Survey.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.3: Balance across the treatment and control group

	Treatment	Control	P-value(Treatment - Control)	Observations
Gender	0.48	0.48	0.994	3130
What is your age?	45.82	46.54	0.211	3130
Non-Hispanic White	0.48	0.45	0.184	3130
African American/Black	0.05	0.06	0.101	3130
Republican	0.26	0.24	0.452	3130
Democrat	0.37	0.38	0.392	3130
Independent	0.36	0.36	0.919	3130
Northeast	0.23	0.22	0.390	3130
Midwest	0.18	0.18	0.968	3130
West	0.23	0.25	0.302	3130
Household size	2.53	2.59	0.245	3130
Log household income	10.91	10.91	0.936	3130
High-skilled	0.76	0.75	0.681	3130
Full-time employee	0.49	0.48	0.530	3130
Part-time employee	0.09	0.08	0.324	3130
Self-employed or small business owner	0.06	0.06	0.295	3130
Unemployed and looking for work	0.05	0.05	0.523	3130
Student	0.03	0.03	0.364	3130
Not in labor force	0.27	0.29	0.124	3130
College	0.85	0.83	0.081	3130
Beliefs about effect of immigration on wages of low-skilled workers	2.20	2.23	0.493	3130
Beliefs about effect of immigration on wages of high-skilled workers	2.96	2.97	0.673	3130
Beliefs about effect of immigration on unemployment of low-skilled workers	2.12	2.08	0.301	3130
Beliefs about effect of immigration on unemployment of high-skilled workers	2.90	2.91	0.890	3130
Confidence in beliefs about wages of low-skilled workers	3.43	3.52	0.024	3130
Confidence in beliefs about unemployment of low-skilled workers	3.50	3.56	0.099	3130
Confidence in beliefsabout wages of high-skilled workers	3.49	3.50	0.657	3130
Confidence in beliefs about unemployment of high-skilled workers	3.44	3.51	0.058	3130

Notes: This table examines covariate balance based on the sample of respondents from the main study. The p-value of an F-test testing for the joint significance of all covariates in predicting treatment status is 0.2302.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.4: Balance across the treatment and control group in the follow-up

	Treatment	Control	P-value(Treatment - Control)	Observations
Gender	0.49	0.50	0.809	3130
What is your age?	46.92	47.97	0.124	3130
Non-Hispanic White	0.48	0.49	0.675	3130
African American/Black	0.05	0.07	0.017	3130
Republican	0.26	0.26	0.891	3130
Democrat	0.37	0.38	0.816	3130
Independent	0.36	0.35	0.791	3130
Northeast	0.23	0.23	0.952	3130
Midwest	0.18	0.20	0.118	3130
West	0.24	0.23	0.462	3130
Household size	2.43	2.48	0.408	3130
Log household income	10.93	10.92	0.760	3130
High-skilled	0.75	0.74	0.734	3130
Full-time employee	0.48	0.45	0.143	3130
Part-time employee	0.09	0.08	0.323	3130
Self-employed or small business owner	0.05	0.06	0.154	3130
Unemployed and looking for work	0.05	0.05	0.980	3130
Student	0.02	0.02	0.442	3130
Not in labor force	0.29	0.33	0.055	3130
College	0.83	0.83	0.855	3130
Beliefs about effect of immigration on wages of low-skilled workers	2.23	2.24	0.729	3130
Beliefs about effect of immigration on wages of high-skilled workers	2.95	2.95	0.974	3130
Beliefs about effect of immigration on unemployment of low-skilled workers	2.12	2.10	0.522	3130
Beliefs about effect of immigration on unemployment of high-skilled workers	2.91	2.91	0.968	3130
Confidence in beliefs about wages of low-skilled workers	3.42	3.51	0.043	3130
Confidence in beliefs about unemployment of low-skilled workers	3.50	3.55	0.212	3130
Confidence in beliefsabout wages of high-skilled workers	3.48	3.50	0.579	3130
Confidence in beliefs about unemployment of high-skilled workers	3.45	3.52	0.077	3130

Notes: This table examines covariate balance for the follow-up sample. The p-value of an F-test testing for the joint significance of all covariates in predicting treatment status is 0.4428.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.5: Attrition in the follow-up study

	Completed Follow-up Survey	Completed Follow-up Survey
Treatment	-0.002	-0.004
	(0.017)	(0.016)
Republican		0.008
		(0.021)
Independent		-0.026
		(0.019)
Log(Income)		-0.020
		(0.011)
High-skilled		-0.114
		(0.020)
Employed Full-Time		-0.010
		(0.023)
Employed Part-Time		0.005
		(0.035)
Unemployed		0.044
		(0.043)
Self-Employed		-0.043
		(0.039)
Student		-0.096
		(0.058)
High Education		-0.045
		(0.032)
Male		0.042
		(0.017)
Age		0.002
		(0.001)
Observations	3130	3130

Notes: The outcome variables takes value 1 for respondents who completed the follow-up study. 'Treatment" is an indicator equal to 1 if respondents received the research evidence. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

C.3 Additional tables

Table A.6: Beliefs about the effects of low-skilled and high-skilled (post-treatment)

	Wages		Employment		Fiscal burden		Culture
	Own household	Most Americans	Own household	Most Americans	Own household	Most Americans	
Panel A: Effect of low-skilled							
Treatment	0.022	0.163	0.015	0.156	0.024	0.030	0.042
	(0.049)	(0.046)	(0.049)	(0.047)	(0.049)	(0.048)	(0.050)
Observations	1474	1474	1469	1469	1464	1464	1452
Panel B: Effect of high-skilled							
Treatment	0.073	0.197	0.155	0.264	0.058	0.081	0.052
	(0.047)	(0.048)	(0.048)	(0.048)	(0.047)	(0.047)	(0.049)
Observations	1476	1476	1470	1470	1464	1464	1449

Notes: The table shows OLS regression results where the dependent variables are beliefs about the economic and cultural impact of different types of immigrants. Respondents were asked how they thought admitting more low-skilled/high-skilled immigrants would affect (i) "wages," (ii) "job opportunities and job security," and (iii) "taxes" for their own household as well as for most Americans. They were also asked about how they thought admitting more low-skilled/high-skilled immigrants "would affect American culture and society as a whole." We randomized whether respondents answered these questions for low-skilled or high-skilled immigrants. All questions were answered on 5-point Likert scales where higher values indicate more optimistic views on the effect of immigration. The outcome variables are z-scored using the mean and standard deviation in the control group. We use the same controls as described in Table 2. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.7: Beliefs about the overall economic impact of immigration (post-treatment)

	Beliefs Low-skilled			Beliefs High-skilled		
	Not familiar	Familiar	Index	Not familiar	Familiar	Index
Panel A: With controls						
Treatment	0.217	0.172	0.195	0.156	0.136	0.144
	(0.048)	(0.050)	(0.043)	(0.048)	(0.048)	(0.044)
Observations	1456	1456	1456	1454	1456	1454
Panel B: Without controls						
Treatment	0.207	0.157	0.182	0.167	0.141	0.153
	(0.052)	(0.052)	(0.047)	(0.050)	(0.049)	(0.045)
Observations	1456	1456	1456	1454	1456	1454

Notes: The table shows OLS regression results where the dependent variables are beliefs about the overall economic impact of the different immigrants. In the main study, respondents were asked how they thought admitting more low-skilled/high-skilled immigrants would affect wages and unemployment for "most Americans." We randomized whether respondents answered these questions for low-skilled or high-skilled immigrants. All questions were answered on 5-point Likert scales where higher values indicate more optimistic views on the effect of immigration. The outcome variables are z-scored using the mean and standard deviation in the control group. We include controls as described in the pre-analysis plan (also listed in Table 2). Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.8: The causal effect of labor market concerns on support for high-skilled immigrants: Subsample of respondents with pessimistic priors

	Labor Market Concerns	Support	Support	
	OLS (reduced form)	OLS (reduced form)	IV	OLS
Panel A: Controls				
Treatment	0.282	0.086		
	(0.044)	(0.033)		
Labor Market Concerns			0.388	0.371
			(0.163)	(0.050)
Observations	1228	2599	1228	600
Panel B: Additional controls				
Treatment	0.144	0.039		
	(0.031)	(0.045)		
Labor Market Concerns			0.273	0.177
			(0.309)	(0.059)
Observations	1211	1211	1211	592

Notes: These regressions are based on the subsample of respondents who thought that the Mariel boatlift had either negative or no effects on wages and unemployment. Column 4 only uses control group respondents, while Colums 1 to 1 use both treatment and control group respondents. The dependent variable in columns 2 to 4 is an index for attitudes towards low-skilled immigrants. The dependent variable in column 1 is "Labor Market Concerns" which is an unweighted index of people's beliefs about the effect of low-skilled immigration on (i) "wages for most Americans" and (ii) "job opportunities or job security for most Americans." In Column 3 we instrument "Labor Market Concerns" with the treatment indicator. In Columns 1, 2 and 4, we run OLS regressions. We randomized whether people were asked about the impact of low-skilled or high-skilled immigrants. All variables have been standardized. All regressions include the controls described in Table 2. Panel B additionally controls for people's beliefs about the impact of immigration on wages and employment of the respondents' own household, as well as beliefs about the tax burden created by immigrants for most Americans and the own household, as well as beliefs about the cultural impact of immigration. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.9: Self-reported attitudes towards immigration (post-treatment) – with pessimistic priors

	Low-skilled (Main Study)			High-skilled (Main Study)			Follow-up	
	Not familiar	Familiar	Index	Not familiar	Familiar	Index	Low-skilled	High-skilled
Panel A: With controls								
Treatment	0.172	0.128	0.150	0.121	0.057	0.089	0.092	0.078
	(0.036)	(0.037)	(0.033)	(0.036)	(0.037)	(0.033)	(0.044)	(0.046)
Adjusted p-value	[0.001]	[0.001]		[0.001]	[0.067]			
Observations	2599	2599	2599	2599	2599	2599	1724	1724
Panel B: Without controls								
Treatment	0.175	0.142	0.159	0.130	0.068	0.099	0.114	0.098
	(0.039)	(0.039)	(0.036)	(0.039)	(0.038)	(0.035)	(0.047)	(0.048)
Adjusted p-value	[0.001]	[0.001]		[0.001]	[0.040]			
Observations	2599	2599	2599	2599	2599	2599	1724	1724

Notes: The table shows OLS regression results where the dependent variables are attitudes to the different types of immigrants. The answers were given on a five point scale from 1: "Allow a lot less of these immigrants" to 5: "Allow a lot more of these immigrants." The question order was randomized (statistical tests show no order effects). The outcomes are z-scored using the mean and standard deviation in the control group. The indices are created by taking the mean of the responses to immigrants with different familiarity with American values and traditions for each skill level. Adjusted p-values are in brackets. "Treatment" is an indicator equal to 1 if respondents received the research evidence. Controls include gender, age, ethnicity, region, household size, household income, education, employment status, party affiliation, whether the respondent was born in the US, whether the subject's parents were born in the US, self-perceived skill-level, and pre-treatment beliefs about the labor market impact of low-skilled (or high-skilled) immigration and are coded as described in the pre-analysis plan. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.10: Self-reported attitudes towards immigration (post-treatment): Sample of respondents who completed the follow-up

	Low-skilled (Main Study)			High-skilled (Main Study)			Follow-up	
	Not familiar	Familiar	Index	Not familiar	Familiar	Index	Low-skilled	High-skilled
Panel A: With controls								
Treatment	0.145	0.104	0.125	0.091	0.043	0.067	0.096	0.064
	(0.040)	(0.042)	(0.037)	(0.041)	(0.042)	(0.037)	(0.040)	(0.042)
Adjusted p-value	[0.001]	[0.006]		[0.051]	[0.183]			
Observations	2075	2075	2075	2075	2075	2075	2075	2075
Panel B: Without controls								
Treatment	0.157	0.119	0.138	0.109	0.052	0.080	0.108	0.070
	(0.044)	(0.044)	(0.040)	(0.043)	(0.043)	(0.039)	(0.044)	(0.043)
Adjusted p-value	[0.001]	[0.003]		[0.024]	[0.129]			
Observations	2075	2075	2075	2075	2075	2075	2075	2075

Notes: The table shows OLS regression results where the dependent variables are attitudes to the different types of immigrants and uses respondents who completed both the initial survey and the follow-up survey. The answers were given on a five point scale from 1: "Allow a lot less of these immigrants" to 5: "Allow a lot more of these immigrants." The question order was randomized (statistical tests show no order effects). The outcomes are z-scored using the mean and standard deviation in the control group. The indices are created by taking the mean of the responses to immigrants with different familiarity with American values and traditions for each skill level. Adjusted p-values are in brackets. "Treatment" is an indicator equal to 1 if respondents received the research evidence. Controls include gender, age, ethnicity, region, household size, household income, education, employment status, party affiliation, whether the respondent was born in the US, whether the subject's parents were born in the US, self-perceived skill-level, and pre-treatment beliefs about the labor market impact of low-skilled (or high-skilled) immigration and are coded as described in the pre-analysis plan. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.11: Correlates of beliefs about the impacts of the Mariel boatlift

	Effects on Low-skilled		Effect on High-skilled	
	Wages	Unemployment	Wages	Unemployment
Republican	-0.226	-0.233	-0.126	-0.161
	(0.046)	(0.046)	(0.047)	(0.047)
Independent	-0.069	-0.050	0.013	0.051
	(0.042)	(0.042)	(0.041)	(0.041)
Age	-0.000	-0.004	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Log(Income)	-0.016	0.009	0.140	0.118
	(0.025)	(0.025)	(0.025)	(0.026)
High-skilled	-0.021	-0.060	-0.169	-0.140
	(0.048)	(0.048)	(0.048)	(0.047)
Employed Full-Time	-0.033	-0.084	-0.169	-0.136
	(0.052)	(0.051)	(0.051)	(0.051)
Employed Part-Time	0.076	0.169	0.032	-0.016
	(0.072)	(0.080)	(0.072)	(0.068)
Unemployed	0.008	0.108	0.040	-0.109
	(0.107)	(0.101)	(0.109)	(0.102)
Self-Employed	-0.014	-0.023	-0.189	-0.101
	(0.083)	(0.090)	(0.084)	(0.082)
Student	0.050	-0.072	0.176	0.251
	(0.126)	(0.124)	(0.139)	(0.144)
High Education	-0.244	-0.112	-0.029	-0.018
	(0.073)	(0.072)	(0.075)	(0.075)
Male	-0.039	-0.019	-0.074	-0.024
	(0.037)	(0.036)	(0.037)	(0.037)
Share of immigrants zip code	0.419	0.249	-0.088	-0.040
	(0.142)	(0.134)	(0.139)	(0.135)
Observations	3130	3130	3130	3130

Notes: The outcome variables are people's pre-treatment beliefs about the effect of the Mariel boatlift on wages and unemployment of low-skilled incumbent workers and high-skilled incumbent workers. All questions were answered on 5-point Likert scales where higher values indicate more optimistic views regarding the effect of immigration. The outcome variables are z-scored using the mean and standard deviation in the control group. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.12: Correlates of attitudes towards immigrants

	Low-skilled		High-skilled	
	Not familiar	Familiar	Not familiar	Familiar
Republican	-0.587	-0.685	-0.430	-0.710
	(0.067)	(0.062)	(0.068)	(0.064)
Independent	-0.220	-0.317	-0.174	-0.310
	(0.055)	(0.055)	(0.056)	(0.055)
Age	-0.009	-0.010	-0.009	-0.010
	(0.002)	(0.002)	(0.002)	(0.002)
Log(Income)	0.061	-0.013	0.062	0.042
	(0.035)	(0.034)	(0.035)	(0.033)
High-skilled	0.157	0.165	0.112	0.103
	(0.064)	(0.064)	(0.065)	(0.063)
Employed Full-Time	0.040	0.069	-0.071	-0.010
	(0.073)	(0.069)	(0.074)	(0.068)
Employed Part-Time	0.071	0.180	-0.054	0.024
	(0.107)	(0.102)	(0.106)	(0.102)
Unemployed	0.027	0.201	-0.096	0.089
	(0.144)	(0.129)	(0.143)	(0.140)
Self-Employed	-0.114	-0.058	-0.208	0.001
	(0.114)	(0.102)	(0.119)	(0.106)
Student	-0.022	0.074	-0.007	0.198
	(0.157)	(0.164)	(0.142)	(0.164)
High Education	-0.020	-0.055	-0.024	-0.064
	(0.097)	(0.094)	(0.097)	(0.094)
Male	-0.049	-0.019	0.117	0.079
	(0.051)	(0.049)	(0.052)	(0.049)
Share of immigrants zip code	-0.128	0.030	-0.181	-0.006
	(0.175)	(0.168)	(0.173)	(0.169)
Observations	1566	1566	1566	1566

Notes: This table uses data from respondents in the control group. The table shows OLS regression results where the dependent variables are attitudes to the four different types of immigrants: (i) low-skilled immigrants not familiar with American values and traditions, (ii) low-skilled immigrants highly familiar with American values and traditions, (iii) high-skilled immigrants not familiar with American values and traditions, and (iv) high-skilled immigrants highly familiar with American values and traditions. The answers were given on a five point scale from 1: "Allow a lot less of these immigrants" to 5: "Allow a lot more of these immigrants." The question order was randomized (statistical tests show no order effects). The outcomes are z-scored using the mean and standard deviation in the control group. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.13: Self-interested labor market concerns vs. group-level labor market concerns

	Low-skilled		High-skilled		Low-skilled		High-skilled	
	Not familiar	Familiar	Not familiar	Familiar	Not familiar	Familiar	Not familiar	Familiar
Effect of immigrants on own household's wage	-0.113	-0.022	-0.057	0.016	-0.137	-0.032	-0.069	0.003
	(0.067)	(0.069)	(0.061)	(0.063)	(0.066)	(0.069)	(0.059)	(0.061)
Effect of immigrants on most Americans' wages	0.207	0.115	0.145	0.139	0.157	0.062	0.062	0.060
	(0.059)	(0.060)	(0.057)	(0.059)	(0.057)	(0.060)	(0.056)	(0.058)
Effect of immigrants on own household's employment	-0.122	-0.108	0.098	0.098	-0.138	-0.122	0.060	0.069
	(0.063)	(0.065)	(0.061)	(0.062)	(0.062)	(0.065)	(0.059)	(0.061)
Effect of immigrants on most Americans' employment	0.406	0.335	0.251	0.185	0.334	0.279	0.202	0.140
	(0.054)	(0.056)	(0.055)	(0.057)	(0.053)	(0.056)	(0.053)	(0.055)
Effect of immigrants on culture					0.257	0.213	0.286	0.267
					(0.036)	(0.037)	(0.037)	(0.038)
Observations	762	762	714	714	750	750	704	704

Notes: This table uses data from respondents in the control group. The table shows OLS regression results where the dependent variables are attitudes to the four different types of immigrants: (i) low-skilled immigrants not familiar with American values and traditions, (ii) low-skilled immigrants highly familiar with American values and traditions, (iii) high-skilled immigrants not familiar with American values and traditions, and (iv) high-skilled immigrants highly familiar with American values and traditions. The answers were given on a five point scale from 1: "Allow a lot less of these immigrants" to 5: "Allow a lot more of these immigrants." The question order was randomized (statistical tests show no order effects). The outcomes are z-scored using the mean and standard deviation in the control group. The independent variables are beliefs about the impact of low-skilled and high-skilled immigrants on the wages and employment of (i) people's own household and (ii) most Americans. On top of that people's beliefs about the effect of immigrants on American culture are included on columns (5) to (8). All questions were answered on 5-point Likert scales where higher values indicate more optimistic views regarding the effect of immigration. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.14: Self-interested vs. nationwide labor market concerns: Evidence from the General Social Survey

	Support for immigration
Belief: Most Americans	0.331
	(0.031)
Belief: own household	0.031
	(0.028)
Observations	1263

Notes: This table uses data from the 1994 wave of the General Social Survey. *Support for immigration* is the standardized response as to whether "the number of immigrants to America nowadays" should be "decreased a lot" to "increased a lot." Self-interested labor market concerns are measured with people's answer to the following question: "What about immigrants? Is it very likely, somewhat likely, somewhat unlikely, or very unlikely that you or anyone in your family won't get a job or promotion while an equally or less qualified immigrant employee receives one instead?" We proxy for people's group-level labor market concerns with people's response to the following question: "How much do you agree or disagree with each of the following statements? Immigrants take jobs away from people who were born in America."

Table A.15: Beliefs about the cultural and overall economic impact of immigration

	Belief:	Belief: Overall Economic Impact	
	Culture	Low-skill	High-skill
Low-skill	-0.329		
	(0.037)		
Not Familiar		-0.400	-0.359
		(0.023)	(0.023)
Control Mean	3.091	3.144	3.243
Observations	2901	1456	1456

Notes: In Column 1 we use the following question: "Setting aside immigration's economic effects, how do you think that increasing the number of **low-skilled** [**high-skilled**] immigrants to the United States would affect American culture and society as a whole?" Respondents answered the question on a scale ranging from (1) "It would greatly damage American culture and society" to (5) "It would greatly improve American culture and society." In columns 2 and 3 we use following questions: "When you think about all of the potential positive and negative economic effects of increasing the number of **high-skilled/low-skilled immigrants** that are **not familiar/highly familiar** with American values and traditions coming to the United States, do you think the overall effect would be positive or negative for the finances of most Americans?" The question was answered on a five point scale from (1) "The overall economic effect would be very negative for most Americans" to (5) "The overall economic effect would be very positive for most Americans." Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

C.4 Heterogeneous effects

Table A.16: Heterogeneous treatment effects by pre-treatment beliefs: Labor market concerns

	Wages		Employment	
	Own household	Most Americans	Own household	Most Americans
Panel A: Effect of low-skilled				
Treatment × Prior: Low skill	-0.102	-0.140	-0.069	-0.179
	(0.062)	(0.059)	(0.058)	(0.056)
Treatment × Prior: High skill	-0.108	-0.071	0.037	0.129
	(0.081)	(0.075)	(0.090)	(0.083)
Treatment	0.573	0.695	0.056	0.155
	(0.269)	(0.238)	(0.288)	(0.263)
Observations	1483	1483	1478	1478
Panel B: Effect of high-skilled				
Treatment × Prior: Low skill	-0.170	-0.188	-0.141	-0.049
	(0.056)	(0.055)	(0.051)	(0.053)
Treatment × Prior: High skill	-0.041	-0.009	-0.041	0.051
	(0.073)	(0.068)	(0.078)	(0.077)
Treatment	0.566	0.629	0.562	0.204
	(0.251)	(0.237)	(0.248)	(0.239)
Observations	1487	1487	1481	1481

Notes: The dependent variables are beliefs about the economic impact of low-skilled and high-skilled immigrants. Respondents were asked how they thought admitting more low-skilled/high-skilled immigrants would affect (i) "wages" and (ii) "job opportunities and job security." We randomized whether respondents answered these questions with respect to low-skilled or high-skilled immigrants. All questions were answered on 5-point Likert scales where higher values indicate more optimistic views regarding the effect of immigration. "Treatment" is an indicator equal to 1 if respondents received the research evidence. "Prior: Low-skill" ("Prior: High-skill") is people's pre-treatment belief about the wage and employment effects of the Mariel boatlift on low-skilled (high-skilled) workers. The regressions include pre-specified controls (listed in Table 2). Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.17: Heterogeneous treatment effects: Self-reports

	Low-skilled				High-skilled			
	Not familiar	Familiar	Index	Follow-up	Not familiar	Familiar	Index	Follow-up
Panel A: Prior Belief								
Treatment ×	-0.067	-0.074	-0.071	-0.039	-0.074	-0.034	-0.054	-0.019
Prior: Low-skilled	(0.043)	(0.045)	(0.040)	(0.051)	(0.044)	(0.045)	(0.040)	(0.056)
Treatment ×	0.068	0.053	0.060	0.003	0.043	0.029	0.036	-0.082
Prior: High-skilled	(0.047)	(0.049)	(0.044)	(0.055)	(0.047)	(0.049)	(0.044)	(0.059)
Treatment	0.169	0.112	0.140	0.096	0.104	0.040	0.072	0.064
	(0.033)	(0.034)	(0.031)	(0.040)	(0.033)	(0.034)	(0.030)	(0.041)
					0.583	0.909	0.712	0.509
Observations	3130	3130	3130	3130	3130	3130	3130	3130
Panel B: Republican								
Treatment × (a)	-0.058	0.002	-0.028	0.055	-0.006	-0.045	-0.025	0.011
Republican	(0.075)	(0.082)	(0.071)	(0.095)	(0.078)	(0.083)	(0.073)	(0.100)
Treatment (b)	0.185	0.111	0.148	0.082	0.106	0.052	0.079	0.061
	(0.038)	(0.038)	(0.035)	(0.045)	(0.038)	(0.038)	(0.034)	(0.046)
Pr(a+b)=0	0.053	0.123	0.055	0.104	0.147	0.925	0.406	0.418
Observations	3130	3130	3130	2075	3130	3130	3130	2075
Panel C: High-skill								
Treatment × (a)	-0.038	-0.062	-0.050	-0.021	0.073	-0.017	0.028	-0.015
High-skill	(0.075)	(0.078)	(0.069)	(0.090)	(0.076)	(0.080)	(0.070)	(0.095)
Treatment (b)	0.199	0.158	0.178	0.112	0.050	0.053	0.051	0.076
	(0.065)	(0.067)	(0.059)	(0.077)	(0.065)	(0.070)	(0.060)	(0.081)
Pr(a+b)=0	0.000	0.016	0.000	0.053	0.002	0.359	0.025	0.215
Observations	3130	3130	3130	2075	3130	3130	3130	2075

Notes: The table shows OLS regression results where the dependent variables are attitudes to the four different types of immigrants: (i) low-skilled immigrants not familiar with American values and traditions, (ii) low-skilled immigrants highly familiar with American values and traditions, (iii) high-skilled immigrants not familiar with American values and traditions, and (iv) high-skilled immigrants highly familiar with American values and traditions. The answers were given on a five point scale from 1: "Allow a lot less of these immigrants" to 5: "Allow a lot more of these immigrants." The question order was randomized (statistical tests show no order effects). The outcomes are z-scored using the mean and standard deviation in the control group. The indices are created by taking the mean of the responses to immigrants with different familiarity with American values and traditions for each skill level. "Treatment" is an indicator equal to 1 if respondents received the research evidence. "Prior: Low-skill" ("Prior: High-skill") is people's pre-treatment belief about the wage and employment effects of the Mariel boatlift on low-skilled (high-skilled) workers. "Republican" takes value 1 if our respondent self-identifies as a Republican and zero otherwise. "High-skill" takes values 1 if our respondent self-identifies as high-skilled and zero otherwise. We use the same controls as described in Table 2. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.18: Heterogeneous treatment effects: Self-reports

	Low-skilled				High-skilled			
	Not familiar	Familiar	Index	Follow-up	Not familiar	Familiar	Index	Follow-up
Panel A: Prior Belief								
Treatment ×	-0.056	-0.063	-0.059	-0.044	0.034	0.026	0.030	-0.079
Prior	(0.043)	(0.045)	(0.040)	(0.051)	(0.046)	(0.049)	(0.044)	(0.057)
Treatment	0.170	0.111	0.141	0.096	0.105	0.040	0.073	0.063
	(0.033)	(0.034)	(0.031)	(0.040)	(0.033)	(0.034)	(0.031)	(0.041)
	0.033	0.383	0.099	0.401	0.011	0.237	0.041	0.811
Observations	3130	3130	3130	2075	3130	3130	3130	2075
Panel B: Republican								
Treatment × (a)	-0.058	0.002	-0.028	0.055	-0.006	-0.045	-0.025	0.011
Republican	(0.075)	(0.082)	(0.071)	(0.095)	(0.078)	(0.083)	(0.073)	(0.100)
Treatment (b)	0.185	0.111	0.148	0.082	0.106	0.052	0.079	0.061
	(0.038)	(0.038)	(0.035)	(0.045)	(0.038)	(0.038)	(0.034)	(0.046)
Pr(a+b)=0	0.053	0.123	0.055	0.104				
Observations	3130	3130	3130	2075	3130	3130	3130	2075
Panel C: High-skill								
Treatment × (a)	-0.038	-0.062	-0.050	-0.021	0.073	-0.017	0.028	-0.015
High-skill	(0.075)	(0.078)	(0.069)	(0.090)	(0.076)	(0.080)	(0.070)	(0.095)
Treatment (b)	0.199	0.158	0.178	0.112	0.050	0.053	0.051	0.076
	(0.065)	(0.067)	(0.059)	(0.077)	(0.065)	(0.070)	(0.060)	(0.081)
Pr(a+b)=0	0.000	0.016	0.000	0.053				
Observations	3130	3130	3130	2075	3130	3130	3130	2075

Notes: The table shows OLS regression results where the dependent variables are attitudes to the four different types of immigrants: (i) low-skilled immigrants not familiar with American values and traditions, (ii) low-skilled immigrants highly familiar with American values and traditions, (iii) high-skilled immigrants not familiar with American values and traditions, and (iv) high-skilled immigrants highly familiar with American values and traditions. The answers were given on a five point scale from 1: "Allow a lot less of these immigrants" to 5: "Allow a lot more of these immigrants." The question order was randomized (statistical tests show no order effects). The outcomes are z-scored using the mean and standard deviation in the control group. The indices are created by taking the mean of the responses to immigrants with different familiarity with American values and traditions for each skill level. "Treatment" is an indicator equal to 1 if respondents received the research evidence. "Prior: Low-skiled" ("Prior: High-skiled") is people's pre-treatment belief about the wage and employment effects of the Mariel boatlift on low-skilled (high-skilled) workers. "Republican" takes value 1 if our respondent self-identifies as a Republican and zero otherwise. "High-skill" takes values 1 if our respondent self-identifies as high-skilled and zero otherwise. We use the same controls as described in Table 2. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.19: Heterogeneous treatment effects: Intention to sign petitions

	Intention: H2B Visas		
	Increase	Decrease	Net support
Panel A: Prior Belief			
Treatment × (a)	-0.004	0.011	-0.027
Prior	(0.019)	(0.019)	(0.042)
Treatment (b)	0.047	-0.058	0.135
	(0.016)	(0.016)	(0.034)
Pr(a+b)=0	0.095	0.041	0.047
Observations	3130	3130	3130
Panel B: Republican			
Treatment × (a)	-0.032	-0.017	-0.014
Republican	(0.033)	(0.038)	(0.076)
Treatment (b)	0.056	-0.054	0.138
	(0.019)	(0.017)	(0.040)
Pr(a+b)=0	0.391	0.039	0.056
Observations	3130	3130	3130
Panel C: High-skill			
Treatment × (a)	-0.010	-0.014	0.018
High-skill	(0.035)	(0.037)	(0.079)
Treatment (b)	0.055	-0.048	0.121
	(0.030)	(0.032)	(0.068)
Pr(a+b)=0	0.016	0.001	0.000
Observations	3130	3130	3130

Notes: The three first columns show regression results where the dependent variable is intention to sign the petitions. "Increase" ("Decrease") is an indicator equal to 1 if a respondent wanted to sign the petition suggesting to increase (decrease) the annual cap on the H-2B visa program. "Net support" is a z-scored transformation of a variable taking value 1 (-1) if a respondent wanted to sign the petition to increase (decrease) the annual cap on the H-2B visa program and 0 otherwise. "Treatment" is an indicator equal to 1 if respondents received the research evidence. "Prior" is people's pre-treatment belief about the wage and employment effects of the Mariel boatlift on low-skilled (high-skilled) workers. "Republican" takes value 1 if our respondent self-identifies as a Republican and zero otherwise. "High-skill" takes values 1 if our respondent self-identifies as high-skilled and zero otherwise. We use the same controls as described in Table 2. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

C.5 Beliefs about the survey and the research evidence

We collected additional data to further understand mechanisms and to assess robustness. To further test for the validity of our evidence, we asked our respondents a series of questions about our experiment, which are summarized in Figure A.9. First, we tested whether our respondents thought that the survey was politically biased. 65 percent of our respondents thought that the survey was not politically biased, 23 percent thought that it was left-wing biased, while the remaining 13 percent perceived the survey as right-wing biased. We also examined whether people's perceptions of bias are correlated with people's responses to the survey, but find only weak correlations with our measures of attitudes towards immigration. People who think that the survey is more right-wing biased display more positive attitudes towards immigration.

We asked all of our respondents whether the Cuban immigrants who came to Miami were mostly high-skilled or low-skilled to test whether our respondents paid attention to the instructions in the belief elicitation. We find that 77.7 percent of respondents correctly remembered that most Cuban immigrants who arrived in Miami were low-skilled.[2] This underlines that most of our respondents actually paid attention to the belief elicitation.

To shed light on mechanisms, we collected rich data among respondents in the treatment group to measure their perception of the research evidence. We first examine to what extent our respondents trusted the information we presented them with. 57 percent of the respondents deemed the research evidence to be "fully trustworthy" or "somewhat trustworthy." 36 percent of the respondents considered the information to be "neither trustworthy nor untrustworthy," while the remaining 7

[2] 76.7 of our respondents thought that most Cuban immigrants who arrived in Miami in 1980 were *not familiar* with American values rather than *highly familiar* with American values.

Electronic copy available at: https://ssrn.com/abstract=3017420

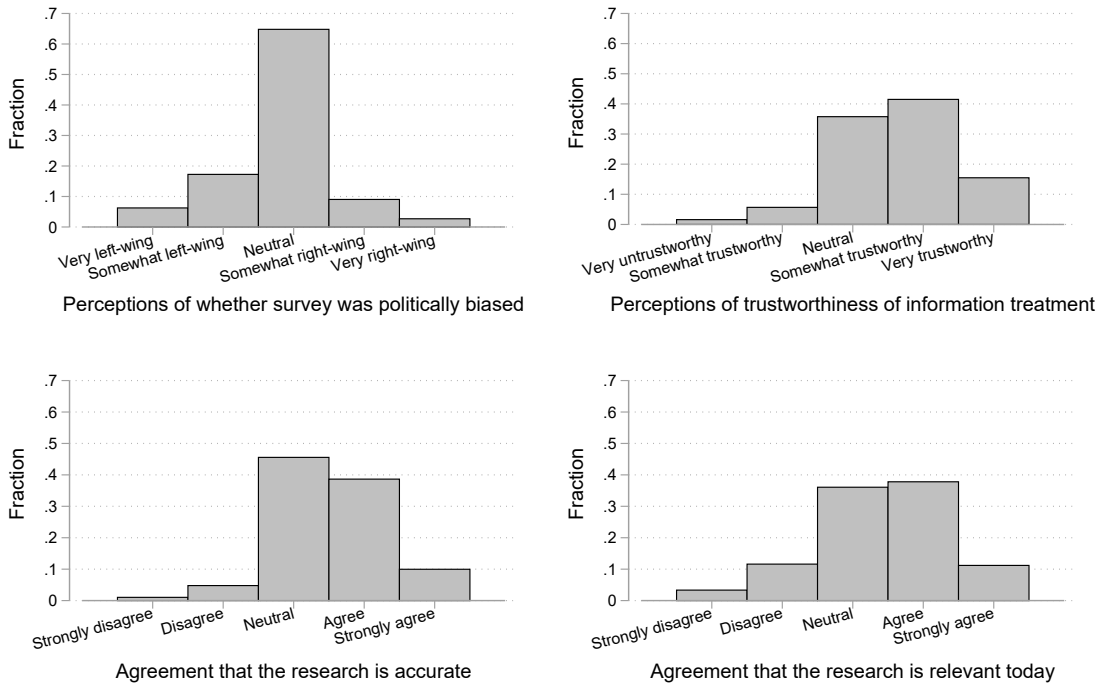
percent of the respondents deemed the research evidence to be "untrustworthy." This suggests that a large fraction of our respondents perceives the research evidence as trustworthy.

49 percent of the respondents "agree" or "strongly agree" that the research evidence about the Mariel boatlift was relevant for "assessing the costs and benefit of allowing more or less immigrants into the US today," while 15 percent "disagree" or "strongly disagree" to this. The remaining 36 percent of the respondents answered that they "neither agree nor disagree" to this. 49 percent of respondents agreed that the research evidence "accurately reflects the labor market effects of the mass immigration of Cubans to Miami in 1980" while 6 percent disagreed with this.

Table A.20 shows that beliefs about the internal and external validity of the study as well as people's beliefs about the trustworthiness of the study are strongly correlated with attitudes towards immigrants in the expected direction: people more skeptical of the research evidence display lower support for immigration conditional on a large set of control variables. Table A.21 examines the correlates of trust in the information provided, as well as beliefs about the internal and external validity of the research evidence provided, among respondents in the treatment group. Republicans, Independents and older respondents display significantly lower levels of trust in the research evidence, lower agreement that the study is externally valid, and that the study is accurate. Education is not correlated with any of these beliefs.

Electronic copy available at: https://ssrn.com/abstract=3017420

Figure A.9: Beliefs about the survey and the research evidence



Notes: The figure displays results on a question measuring people's perception of the trustworthiness in the research evidence. The figure shows the fraction of respondents who think that the survey was (1) very untrustworthy, (2) untrustworthy, (3) neither untrustworthy nor trustworthy, (4) trustworthy, and (5) very trustworthy.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.20: Beliefs about the research study and attitudes towards immigrants

	Index: Attitudes towards	
	low-skilled immigrants	high-skilled immigrants
Info: trustworthy	0.179	0.168
	(0.036)	(0.037)
Study: internally valid	0.069	0.059
	(0.040)	(0.041)
Study: externally valid	0.175	0.147
	(0.028)	(0.029)
Observations	1439	1439

Notes: This table uses data from respondents in the treatment group. The table shows OLS regression results where the dependent variables are indices of attitudes towards lows-skilled and high-skilled immigrants. The index on low-skilled immigrants is based on the following two types of immigrants: (i) low-skilled immigrants not familiar with American values and traditions. and (ii) low-skilled immigrants highly familiar with American values and traditions. The index on high-skilled immigrants is based on the following two types of immigrants: (i) high-skilled immigrants not familiar with American values and traditions, and (ii) high-skilled immigrants highly familiar with American values and traditions. The answers were given on a five point scale from 1: "Allow a lot less of these immigrants" to 5: "Allow a lot more of these immigrants." The question order was randomized (statistical tests show no order effects). The outcomes are z-scored using the mean and standard deviation in the control group. The main independent variables in these regressions are people's assessment of whether the research evidence we provided them with is (i) trustworthy, (ii) reflective of the actual labor market impacts of the Mariel boatlift, and (iii) relevant for assessing the costs and benefits of allowing more or less immigrants into the US today. The questions were asked on 5-point Likert scales, where higher values indicate more positive views on the research evidence. We use the same controls as described in Table 2. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

Table A.21: Correlates of beliefs about the research study

	The information is	The research study is	
	trustworthy	internally valid	externally valid
Republican	-0.389	-0.358	-0.482
	(0.059)	(0.053)	(0.065)
Independent	-0.308	-0.269	-0.360
	(0.052)	(0.046)	(0.055)
Age	-0.004	-0.005	-0.007
	(0.002)	(0.002)	(0.002)
Log(Income)	-0.001	-0.012	0.011
	(0.030)	(0.028)	(0.033)
High-skilled	0.094	0.080	0.054
	(0.059)	(0.053)	(0.062)
Employed Full-Time	-0.122	-0.039	-0.063
	(0.063)	(0.056)	(0.069)
Employed Part-Time	-0.125	-0.119	-0.059
	(0.089)	(0.080)	(0.092)
Unemployed	-0.233	-0.119	-0.037
	(0.115)	(0.116)	(0.123)
Self-Employed	-0.200	-0.175	-0.260
	(0.123)	(0.105)	(0.124)
Student	-0.045	-0.030	0.067
	(0.155)	(0.155)	(0.156)
High Education	-0.038	-0.017	-0.015
	(0.087)	(0.079)	(0.090)
Male	0.015	0.040	0.022
	(0.046)	(0.041)	(0.050)
Observations	1446	1444	1439

Notes: This table uses data from respondents in the treatment group. The main dependent variables in this regressions are people's assessment of whether the research evidence we provided them with is (i) trustworthy, (ii) reflective of the actual labor market impacts of the Mariel boatlift, and (iii) relevant for assessing the costs and benefits of allowing more or less immigrants into the US today. The questions were asked on 5-point Likert scales, where higher values indicate more positive views on the research evidence. Robust standard errors in parentheses.

Electronic copy available at: https://ssrn.com/abstract=3017420

D Miscellaneous

Figure A.10: Petition

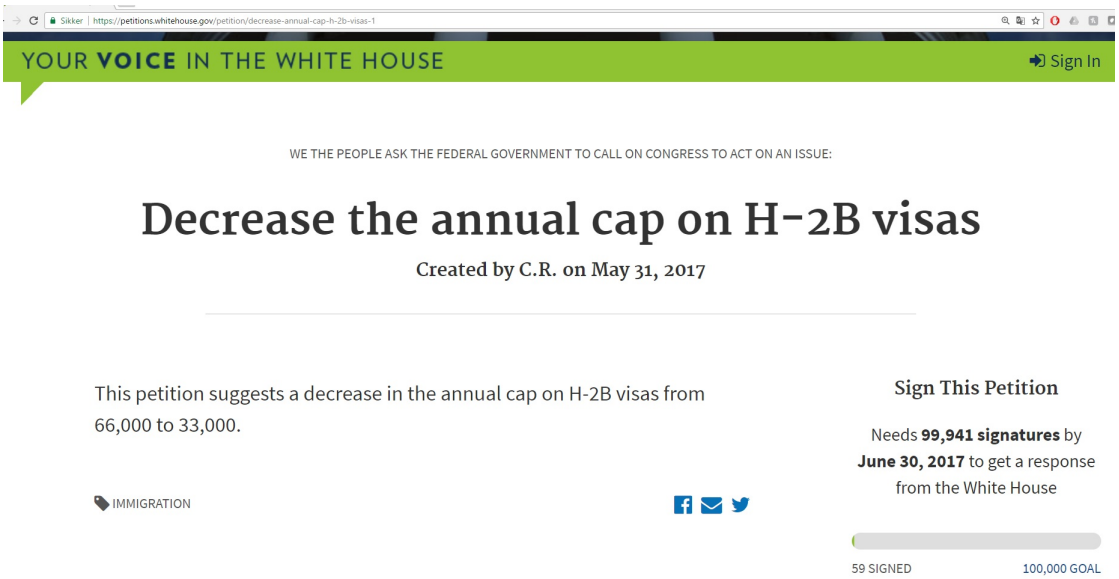
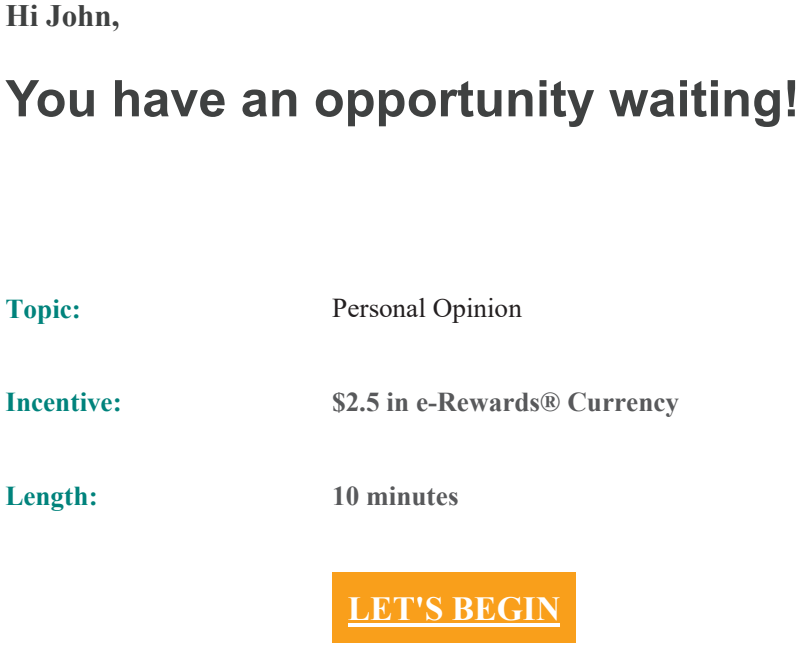
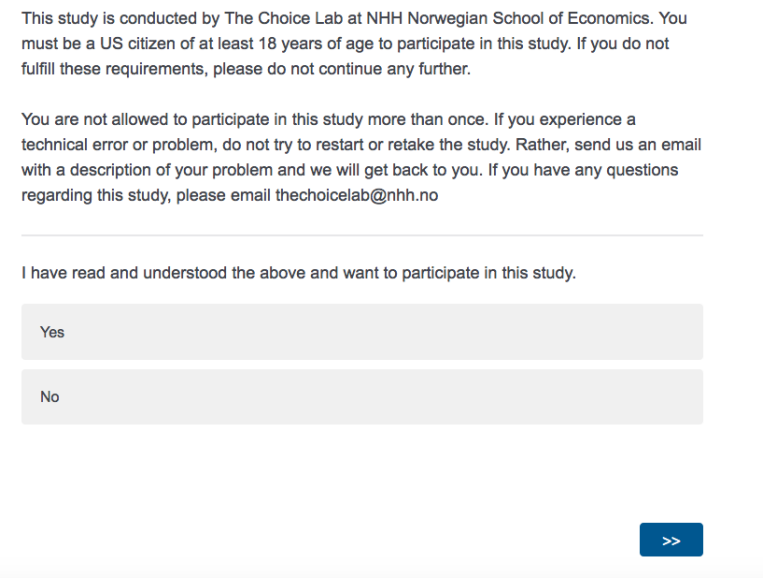


Figure A.11: Invitation form in the email sent out for the obfuscated follow-up study



Electronic copy available at: https://ssrn.com/abstract=3017420

Figure A.12: Consent form in the main study



This study is conducted by The Choice Lab at NHH Norwegian School of Economics. You must be a US citizen of at least 18 years of age to participate in this study. If you do not fulfill these requirements, please do not continue any further.

You are not allowed to participate in this study more than once. If you experience a technical error or problem, do not try to restart or retake the study. Rather, send us an email with a description of your problem and we will get back to you. If you have any questions regarding this study, please email thechoicelab@nhh.no

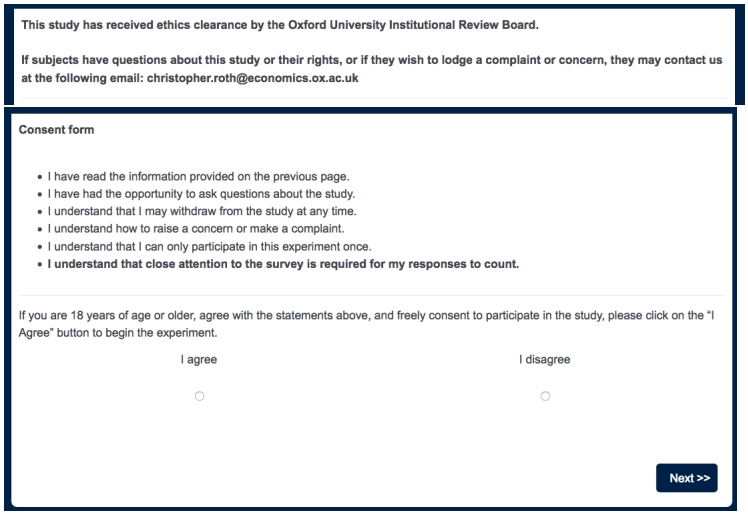
I have read and understood the above and want to participate in this study.

Yes

No

>>

Figure A.13: Consent form in the follow-up study



This study has received ethics clearance by the Oxford University Institutional Review Board.

If subjects have questions about this study or their rights, or if they wish to lodge a complaint or concern, they may contact us at the following email: christopher.roth@economics.ox.ac.uk

Consent form

- I have read the information provided on the previous page.
- I have had the opportunity to ask questions about the study.
- I understand that I may withdraw from the study at any time.
- I understand how to raise a concern or make a complaint.
- I understand that I can only participate in this experiment once.
- **I understand that close attention to the survey is required for my responses to count.**

If you are 18 years of age or older, agree with the statements above, and freely consent to participate in the study, please click on the "I Agree" button to begin the experiment.

I agree

I disagree

Next >>

Figure A.14: Screenshot of the informational treatment

The researchers who analyzed the short- and long-term effects of the mass immigration of Cubans to Miami concluded that, for both high-skilled and low-skilled workers, the mass immigration had **virtually no effect on wages** and **virtually no effect on unemployment**.

According to the researchers, the mass immigration had virtually no effect on wages and unemployment because the new Cuban immigrants increased the overall demand for goods and services, which created more jobs.

Electronic copy available at: https://ssrn.com/abstract=3017420

For online publication only:

Labor Market Concerns and Support for Immigration: Experimental instructions

Ingar Haaland and Christopher Roth

E Instructions: Main experiment

E.1 Introduction

General instructions

This study is conducted by The Choice Lab at NHH Norwegian School of Economics.

You must be a US citizen of at least 18 years of age to participate in this study. If you do not fulfill these requirements, please do not continue any further.

You are not allowed to participate in this study more than once. If you experience a technical error or problem, do not try to restart or retake the study. Rather, send us an email with a description of your problem and we will get back to you.

If you have any questions regarding this study, please email thechoicelab@nhh.no.

I have read and understood the above and want to participate in this study.

Yes

No

Electronic copy available at: https://ssrn.com/abstract=3017420

E.2 Attention check

The next question is about the following problem. In questionnaires like ours, sometimes there are subjects who do not carefully read the questions and just quickly click through the survey. This means that there are a lot of random answers which compromise the results of research studies. To show that you read our questions carefully, please choose "Extremely interested" and "Not at all interested" as your answer in the next question.

How interested are you in sports?

Extremely interested

Very interested

A little bit interested

Almost not interested

Not at all interested

E.3 Pre-treatment background questions

1. What is your age? [18–24; 25–34; 35–44; 45–54; 55–64; 65 or older]

2. What is your gender? [Male; Female]

3. What was your family's gross household income in 2016 in US dollars? [Less than $15,000; $15,000 to $24,999; $25,000 to $49,999; $50,000 to $74,999; $75,000 to $99,999; $100,000 to $149,999; $150,000 to $200,000; More than $200,000; Prefer not to answer]

Electronic copy available at: https://ssrn.com/abstract=3017420

4. Which of the following best describes your race or ethnicity? [African American/Black; Asian/Asian American; Caucasian/White; Native American, Inuit or Aleut; Native Hawaiian/Pacific Islander; Other; Prefer not to answer]

5. Are you of Hispanic, Latino, or Spanish origin? [Yes, No]

6. In which state do you currently reside?

7. In politics, as of today, do you consider yourself a Republican, a Democrat, or an Independent? [Republican, Democrat, Independent]

E.4 Self-perceived skill level

People in the workforce differ by their professional skill levels. A high-skilled worker is someone who is highly educated or has special training and knowledge. A low-skilled worker is someone who does not have extensive education or special training or knowledge. A high-skilled worker is someone like an engineer or doctor, who is highly educated, or a computer programmer with special knowledge. A low-skilled worker is someone like an agricultural worker, housekeeper, or laborer who does not have extensive education or special knowledge.

In this context, do you consider yourself to be low-skilled or high-skilled?

High-skilled

Low-skilled

E.5 Elicitation of beliefs: labor market

We will now ask you a series of questions about the labor market impacts of immigrants. **It is important that you read all of the text carefully before you submit**

Electronic copy available at: https://ssrn.com/abstract=3017420

your answers.

{*page break*}

Background

In 1980, Cuba's then President, Fidel Castro, suddenly announced that Cubans wishing to emigrate to the United States were free to do so. This led to an unexpected mass immigration to Miami, Florida, where most of the Cuban immigrants arrived by boat.

With the arrival of the new Cuban immigrants, Miami's workforce grew by 55,000, or 8 percent, almost at once. The new immigrants were mostly low-skilled, which meant that the low-skilled workforce increased by 20 percent.

The large, unexpected addition of 55,000 new immigrants to the Miami workforce has allowed researchers to study the impact of immigration on the labor market. To do so, the researchers studied wage and unemployment changes in Miami after the mass immigration relative to other US cities that, because of geographic distance, were not affected by the mass immigration of Cubans.

What do you think?

In the five-year period after 1980, how do you think wages of **low-skilled** workers in Miami were affected by the mass immigration of Cubans?

> I think the mass immigration of Cuban workers strongly decreased wages of low-skilled workers
>
> I think the mass immigration of Cuban workers somewhat decreased wages of low-skilled workers
>
> I think the mass immigration had virtually no effect on wages of low-skilled workers

Electronic copy available at: https://ssrn.com/abstract=3017420

I think the mass immigration of Cuban workers somewhat increased wages of low-skilled workers

I think the mass immigration of Cuban workers strongly increased wages of low-skilled workers

{*page break*}

How sure are you about your answer to the previous question?

Very sure

Sure

Somewhat sure

Unsure

Very unsure

{*page break*}

In the five-year period after 1980, how do you think wages of **high-skilled** workers in Miami were affected by the mass immigration of Cubans?

I think the mass immigration of Cuban workers strongly decreased wages of high-skilled workers

I think the mass immigration of Cuban workers somewhat decreased wages of high-skilled workers

I think the mass immigration had virtually no effect on wages of high-skilled workers

Electronic copy available at: https://ssrn.com/abstract=3017420

I think the mass immigration of Cuban workers somewhat increased wages of high-skilled workers

I think the mass immigration of Cuban workers strongly increased wages of high-skilled workers

{*page break*}

How sure are you about your answer to the previous question?

Very sure

Sure

Somewhat sure

Unsure

Very unsure

{*page break*}

In the five-year period after 1980, how do you think unemployment among **low-skilled** workers in Miami was affected by the mass immigration of Cubans?

I think the mass immigration of Cuban workers strongly increased unemployment among low-skilled workers

I think the mass immigration of Cuban workers somewhat increased unemployment among low-skilled workers

I think the mass immigration had virtually no effect on unemployment for low-skilled workers

Electronic copy available at: https://ssrn.com/abstract=3017420

I think the mass immigration of Cuban workers somewhat decreased unemployment among low-skilled workers

I think the mass immigration of Cuban workers strongly decreased unemployment among low-skilled workers

{*page break*}

How sure are you about your answer to the previous question?

Very sure

Sure

Somewhat sure

Unsure

Very unsure

{*page break*}

In the five-year period after 1980, how do you think unemployment among **high-skilled** workers in Miami was affected by the mass immigration of Cubans?

I think the mass immigration of Cuban workers strongly increased unemployment among high-skilled workers

I think the mass immigration of Cuban workers somewhat increased unemployment among high-skilled workers

I think the mass immigration had virtually no effect on unemployment for high-skilled workers

Electronic copy available at: https://ssrn.com/abstract=3017420

I think the mass immigration of Cuban workers somewhat decreased unemployment among high-skilled workers

I think the mass immigration of Cuban workers strongly decreased unemployment among high-skilled workers

{*page break*}

How sure are you about your answer to the previous question?

Very sure

Sure

Somewhat sure

Unsure

Very unsure

E.6 Treatment: Labor market impacts

The researchers who analyzed the short- and long-term effects of the mass immigration of Cubans to Miami concluded that, for both high-skilled and low-skilled workers, the mass immigration had **virtually no effect on wages** and **virtually no effect on unemployment**.

According to the researchers, the mass immigration had virtually no effect on wages and unemployment because the new Cuban immigrants increased the overall demand for goods and services, which created more jobs.

Electronic copy available at: https://ssrn.com/abstract=3017420

E.7 Self-reported outcomes

E.7.1 Attitudes toward immigration: randomized order

Immigrants to the US differ in terms of their professional skill levels as well as their familiarity with American values and traditions.

Do you think the US should allow more or less **low-skilled** immigrants that are **highly familiar** with American values and traditions to come and live here?

Allow a lot more of these immigrants

Allow more of these immigrants

Keep the numbers as they are

Allow less of these immigrants

Allow a lot less of these immigrants

{*page break*}

Do you think the US should allow more or less **high-skilled** immigrants that are **highly familiar** with American values and traditions to come and live here?

Allow a lot more of these immigrants

Allow more of these immigrants

Keep the numbers as they are

Allow less of these immigrants

Allow a lot less of these immigrants

Electronic copy available at: https://ssrn.com/abstract=3017420

{*page break*}

Do you think the US should allow more or less **low-skilled** immigrants that are **not familiar** with American values and traditions to come and live here?

Allow a lot more of these immigrants

Allow more of these immigrants

Keep the numbers as they are

Allow less of these immigrants

Allow a lot less of these immigrants

{*page break*}

Do you think the US should allow more or less **high-skilled** immigrants that are **not familiar** with American values and traditions to come and live here?

Allow a lot more of these immigrants

Allow more of these immigrants

Keep the numbers as they are

Allow less of these immigrants

Allow a lot less of these immigrants

E.8 Petition

H-2B visas are work permits that allow US companies to temporarily hire low-skilled workers from abroad for seasonal, non-agricultural jobs, typically for work

Electronic copy available at: https://ssrn.com/abstract=3017420

in restaurants, tourism, or construction. The annual cap on H-2B visas is currently 66,000. Congress is debating whether to change the annual cap.

Some argue that the quota should be increased because private companies say that there are not enough low-skilled American workers for hire. Others argue that the quota should be decreased because access to more foreign workers makes it easier for private companies to cut the wages of low-skilled American workers.

You will now have the possibility of signing a real petition related to this debate. If enough people sign the petition, the White House will consider it and post an official response.

Consider the following two petitions and decide whether you would like to sign one of them:

Increase the annual cap on H-2B visas (randomized order)

This petition suggests an increase in the annual cap on H-2B visas from 66,000 to 99,000.

I want to sign this petition.

I do not want to sign this petition.

Decrease the annual cap on H-2B visas (randomized order)

This petition suggests a decrease in the annual cap on H-2B visas from 66,000 to 33,000.

I want to sign this petition.

I do not want to sign this petition.

{*page break*}

Electronic copy available at: https://ssrn.com/abstract=3017420

(Not shown if "I do not want to sign this petition" was selected for both of the previous questions)

You stated that you want to sign the petition **"{*petition name*}"**. If you are interested in signing the petition, please click on the link below.

`https://petitions.whitehouse.gov/petition/...`

E.9 Beliefs about the effects of low-skilled immigration

Note: We randomize between subjects whether they are exposed to this block about low-skilled immigration or the next block about high-skilled immigration.

For the following questions, we would like you to think about how increasing the number of **low-skilled** immigrants to the United States would affect your household and how it would affect most Americans over the next five years.

E.9.1 Labor market prospects: wages

For **your household**, how do you think admitting more **low-skilled** immigrants will affect wages?

Strongly decrease wages for my household

Somewhat decrease wages for my household

Neither increase nor decrease wages for my household

Somewhat increase wages for my household

Strongly increase wages for my household

Electronic copy available at: https://ssrn.com/abstract=3017420

For **most Americans**, how do you think admitting more **low-skilled** immigrants will affect wages?

Strongly decrease wages for most Americans

Somewhat decrease wages for most Americans

Neither increase nor decrease wages for most Americans

Somewhat increase wages for most Americans

Strongly increase wages for most Americans

E.9.2 Labor market prospects: jobs

For **your household**, how do you think admitting more **low-skilled** immigrants will affect job opportunities and job security?

Strongly reduce job opportunities or job security for my household

Somewhat reduce job opportunities or job security for my household

Neither reduce nor increase job opportunities or job security for my household

Somewhat increase job opportunities or job security for my household

Strongly increase job opportunities or job security for my household

For **most Americans**, how do you think admitting more **low-skilled** immigrants will affect job opportunities and job security?

Strongly reduce job opportunities or job security for most Americans

Electronic copy available at: https://ssrn.com/abstract=3017420

Somewhat reduce job opportunities or job security for most Americans

Neither reduce nor increase job opportunities or job security for most Americans

Somewhat increase job opportunities or job security for most Americans

Strongly increase job opportunities or job security for most Americans

E.9.3 Fiscal burden

For **your household**, how do you think admitting more **low-skilled** immigrants will affect taxes?

Increase taxes a lot for my household

Increase taxes a little for my household

Have no effect on taxes for my household

Decrease taxes a little for my household

Decrease taxes a lot for my household

For **most Americans**, how do you think admitting more **low-skilled** immigrants will affect taxes?

Increase taxes a lot for most Americans

Increase taxes a little for most Americans

Have no effect on taxes for most Americans

Decrease taxes a little for most Americans

Decrease taxes a lot for most Americans

Electronic copy available at: https://ssrn.com/abstract=3017420

E.9.4 Overall economic impact

When you think about all of the potential positive and negative economic effects of increasing the number of **low-skilled** immigrants that are **highly familiar** with American values and traditions coming to the United States, do you think the overall effect would be positive or negative for the finances of **most Americans**?

The overall economic effect would be very positive for most Americans

The overall economic effect would be somewhat positive for most Americans

There would be no economic effect for most Americans

The overall economic effect would be somewhat negative for most Americans

The overall economic effect would be very negative for most Americans

{*page break*}

When you think about all of the potential positive and negative economic effects of increasing the number of **low-skilled** immigrants that are **not familiar** with American values and traditions coming to the United States, do you think the overall effect would be positive or negative for the finances of **most Americans**?

The overall economic effect would be very positive for most Americans

The overall economic effect would be somewhat positive for most Americans

There would be no economic effect for most Americans

The overall economic effect would be somewhat negative for most Americans

The overall economic effect would be very negative for most Americans

Electronic copy available at: https://ssrn.com/abstract=3017420

{*page break*}

Setting aside immigration's economic effects, how do you think that increasing the number of **low-skilled** immigrants to the United States would affect American culture and society as a whole?

It would greatly damage American culture and society

It would somewhat damage American culture and society

It would neither damage nor improve American culture and society

It would somewhat improve American culture and society

It would greatly improve American culture and society

E.10 Beliefs about the effects of high-skilled immigration

For the following questions, we would like you to think about how increasing the number of **high-skilled** immigrants to the United States would affect your household and how it would affect most Americans over the next five years.

E.10.1 Labor market prospects: wages

For **your household**, how do you think admitting more **high-skilled** immigrants will affect wages?

Strongly decrease wages for my household

Somewhat decrease wages for my household

Neither increase nor decrease wages for my household

Electronic copy available at: https://ssrn.com/abstract=3017420

Somewhat increase wages for my household

Strongly increase wages for my household

For **most Americans**, how do you think admitting more **high-skilled** immigrants will affect wages?

Strongly decrease wages for most Americans

Somewhat decrease wages for most Americans

Neither increase nor decrease wages for most Americans

Somewhat increase wages for most Americans

Strongly increase wages for most Americans

E.10.2 Labor market prospects: jobs

For **your household**, how do you think admitting more **high-skilled** immigrants will affect job opportunities and job security?

Strongly reduce job opportunities or reduce job security for my household

Somewhat reduce job opportunities or reduce job security for my household

Neither reduce nor increase job opportunities and job security for my household

Somewhat increase job opportunities or increase job security for my household

Strongly increase job opportunities or increase job security for my household

Electronic copy available at: https://ssrn.com/abstract=3017420

For **most Americans**, how do you think admitting more **high-skilled** immigrants will affect job opportunities and job security?

Strongly reduce job opportunities or reduce job security for most Americans

Somewhat reduce job opportunities or reduce job security for most Americans

Neither reduce nor increase job opportunities and job security for most Americans

Somewhat increase job opportunities or increase job security for most Americans

Strongly increase job opportunities or increase job security for most Americans

E.10.3 Fiscal burden

For **your household**, how do you think admitting more **high-skilled** immigrants will affect taxes?

Increase taxes a lot for my household

Increase taxes a little for my household

Have no effect on taxes for my household

Decrease taxes a little for my household

Decrease taxes a lot for my household

For **most Americans**, how do you think admitting more **high-skilled** immigrants will affect taxes?

Electronic copy available at: https://ssrn.com/abstract=3017420

Increase taxes a lot for most Americans

Increase taxes a little for most Americans

Have no effect on taxes for most Americans

Decrease taxes a little for most Americans

Decrease taxes a lot for most Americans

E.10.4 Overall economic impact

When you think about all of the potential positive and negative economic effects of increasing the number of **high-skilled** immigrants that are **highly familiar** with American values and traditions coming to the United States, do you think the overall effect would be positive or negative for the finances of **most Americans**?

The overall economic effect would be very positive for most Americans

The overall economic effect would be somewhat positive for most Americans

There would be no economic effect for most Americans

The overall economic effect would be somewhat negative for most Americans

The overall economic effect would be very negative for most Americans

{*page break*}

When you think about all of the potential positive and negative economic effects of increasing the number of **high-skilled** immigrants that are **not familiar** with American values and traditions coming to the United States, do you think the overall effect would be positive or negative for the finances of **most Americans**?

Electronic copy available at: https://ssrn.com/abstract=3017420

The overall economic effect would be very positive for most Americans

The overall economic effect would be somewhat positive for most Americans

There would be no economic effect for most Americans

The overall economic effect would be somewhat negative for most Americans

The overall economic effect would be very negative for most Americans

{*page break*}

Setting aside immigration's economic effects, how do you think that increasing the number of **high-skilled** immigrants to the United States would affect American culture and society as a whole?

It would greatly damage American culture and society

It would somewhat damage American culture and society

It would neither damage nor improve American culture and society

It would somewhat improve American culture and society

It would greatly improve American culture and society

E.11 Questions about the study

E.11.1 Questions asked to all subjects

Do you feel this survey was politically biased?

Very left-wing biased

Electronic copy available at: https://ssrn.com/abstract=3017420

Somewhat left-wing biased

Neither left-wing nor right-wing biased

Somewhat right-wing biased

Very right-wing biased

{*page break*}

Previously in this survey, we gave you some information about the mass immigration of Cubans to Miami in 1980. Please answer the following question to the best of your memory.

Were most of the Cuban immigrants that came to Miami in 1980 high-skilled or low-skilled?

High-skilled

Low-skilled

I am unsure

{*page break*}

Do you think most of the Cubans who migrated to Miami in 1980 were **highly familiar** or **not familiar** with American values and traditions?

Highly familiar

Not familiar

Electronic copy available at: https://ssrn.com/abstract=3017420

E.11.2 Questions only asked to subjects in the treatment

We provided you with information about the results from research on the labor market consequences of the mass immigration of Cubans to Miami in 1980. Did you find the information we provided you with trustworthy or untrustworthy?

Very trustworthy

Somewhat trustworthy

Neither trustworthy nor untrustworthy

Somewhat untrustworthy

Very untrustworthy

To what extent to do you agree with the following statement: "The research described in this survey accurately reflects the labor market effects of the mass immigration of Cubans to Miami in 1980."

Strongly disagree

Disagree

Neither agree nor disagree

Agree

Strongly agree

To what extent do you agree with the following statement: "The research on how the mass immigration of Cubans to Miami in 1980 affected the labor market is relevant when assessing the costs and benefits of allowing more or less immigrants into the United States today."

Electronic copy available at: https://ssrn.com/abstract=3017420

Strongly Disagree

Disagree

Neither agree nor disagree

Agree

Strongly Agree

E.12 Demographics

1. Including yourself, how many people are currently living in your household?
2. Which category best describes your highest level of education? [Eighth grade or less, Some high school, High school degree/GED, Some college, 2-year college degree, 4-year college degree, Master's degree, Doctoral degree, Professional degree (JD, MD, MBA)]
3. What is your current employment status? [Full-time employee, Part-time employee, Self-employed or small business owner, Unemployed and looking for work, Student, Not in labor force (for example: retired or full-time parent)]
4. What is the zip code of your current residence?
5. Were you born in the US? [Yes, No]
6. Were both of your parents born in the US? [Yes, No]

E.13 Questions about employment

(Not shown if any of "Unemployed and looking for work", "Student", or "Not in labor force" were selected for the question about current employment status.)

Electronic copy available at: https://ssrn.com/abstract=3017420

Which of the following categories best describes the industry you primarily work in (regardless of your actual position)?

- ◯ Computer programmers
- ◯ Telemarketers
- ◯ Computer systems analysts
- ◯ Billing and posting clerks and Machine operators
- ◯ Bookkeeping, accounting, And auditing clerks
- ◯ Computer support specialists
- ◯ Computer software engineers, Applications
- ◯ Computer software engineers, systems software
- ◯ Accountants
- ◯ Welders, cutters, solderers, and brazers
- ◯ Helpers–production workers
- ◯ First-line supervisors/managers of production and operating workers
- ◯ Packaging and filling machine operators and tenders
- ◯ Team assemblers
- ◯ Bill and account collectors
- ◯ Machinists
- ◯ Inspectors, testers, sorters, samplers, and weighers
- ◯ General and operations managers
- ◯ Stock clerks and order fillers
- ◯ Shipping, receiving, and traffic clerks
- ◯ Sales managers
- ◯ Business operations specialists, all other
- ◯ Does not apply.

{*page break*}

(Not shown if "Does not apply" was selected for the previous questions)

What do you think is the share of immigrants working in your local industry? [0 percent, ... , 100 percent]

Electronic copy available at: https://ssrn.com/abstract=3017420

F Instructions: Follow-up

F.1 Consent form

This study has received ethics clearance by the Oxford University Institutional Review Board.

If subjects have questions about this study or their rights, or if they wish to lodge a complaint or concern, they may contact us at the following email: Christopher.Roth@economics.ox.ac.uk

{*page break*}

Consent form

- I have read the information provided on the previous page.
- I have had the opportunity to ask questions about the study.
- I understand that I may withdraw from the study at any time.
- I understand how to raise a concern or make a complaint.
- I understand that I can only participate in this experiment once.
- **I understand that close attention to the survey is required for my responses to count.**

If you are 18 years of age or older, agree with the statements above, and freely consent to participate in the study, please click on the "I Agree" button to begin the experiment. [I agree/I disagree]

Electronic copy available at: https://ssrn.com/abstract=3017420

F.2 Obfuscation: Demographics

1. What is your sex? [Male, Female, Prefer not to answer]

2. How old are you (in years)? [18–30, 30–50, 50–70, Older than 70, Prefer not to answer]

3. Information about income is very important to understand. Would you please give your best guess? Please indicate the answer that includes your entire household income in 2016 before taxes. [Income brackets from "Less than $10 000" to "Less than $150 000 or more"; Prefer not to answer]

4. Which statement best describes your current employment status? [Working (paid employee), Working (self-employed), Not working (looking for work), Not working (retired), Not working (temporary layoff from a job), Not working (disabled), Not working (other), Prefer not to answer]

5. Do you consider yourself a Republican, a Democrat, or an Independent? [Republican, Democrat, Independent, Prefer not to answer]

F.3 Obfuscation: Views on government

Do you think the overall amount of government spending should be increased, decreased, or remain the same?

Strongly increased

Somewhat increased

Kept at its present level

Electronic copy available at: https://ssrn.com/abstract=3017420

Somewhat decreased

Strongly decreased

{*page break*}

Do you think the overall amount of taxes raised by the government should be increased, decreased, or remain the same?

Strongly increased

Somewhat increased

Kept at its present level

Somewhat decreased

Strongly decreased

{*page break*}

People feel differently about how far a government should go. Here is a phrase which some people believe in and some don't. Do you think the government should or should not redistribute wealth by heavy taxes on the rich?

Yes, redistribute by heavy taxes on rich

No, should not redistribute wealth

No opinion

Electronic copy available at: https://ssrn.com/abstract=3017420

F.4 Key outcomes

We would like you to think about immigrants – that is, people who come from other countries to live here in the United States. In your view, should immigration of workers with little to no education be kept at its present level, increased, or decreased?

Strongly increased

Somewhat increased

Kept at its present level

Somewhat decreased

Strongly decreased

{*page break*}

In your view, should immigration of highly educated workers be kept at its present level, increased, or decreased?

Strongly increased

Somewhat increased

Kept at its present level

Somewhat decreased

Strongly decreased

Electronic copy available at: https://ssrn.com/abstract=3017420

{*page break*}

To what extent do you agree with the following statement: “Increased immigration hurts American workers”

Strongly agree

Agree

Neither agree nor disagree

Disagree

Strongly disagree

Electronic copy available at: https://ssrn.com/abstract=3017420